

# The Conjectures and Questions of Kozma and Nitzan on Bernoulli Percolation

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## Abstract

Kozma and Nitzan conjectured inequalities comparing the probability that one vertex of a finite weighted graph is joined to another with the probabilities that a prescribed finite set of vertices is joined to each of them, and they proved that the weakest of these implies that critical Bernoulli bond percolation on the hypercubic lattice has no infinite open cluster in every dimension at least two. Their paper contains nine numbered statements. Eight of them are proved here, most in a form stronger than the printed one, and the ninth is answered negatively in the reading that allows a tie. The whole argument rests on one inequality, which bounds the mean of an increasing function of the open cluster of a vertex from below by the mean of the same function at the first vertex of the prescribed set that the cluster reaches, and which is proved by removing the vertices of that set one at a time. Two features of it carry all the strength: the whole inequality may be conditioned on the cluster missing an auxiliary set of vertices, and a finite set of vertices may replace the single starting vertex, which is what turns it into a statement about a graph with one edge forced open.

## 1 Introduction

Let every unordered pair of vertices of a finite vertex set be open independently with a prescribed probability, and write  $x \leftrightarrow y$  when the two vertices are joined by a path of open pairs. Kozma and Nitzan asked how the probability that a vertex  $o$  is joined to a vertex  $b$  compares with the probabilities that  $o$  reaches a prescribed finite set  $A$  and that the vertices of  $A$  reach  $b$ . A path from  $o$  to  $A$  and a path from a vertex of  $A$  to  $b$  should combine into a path from  $o$  to  $b$ , but the two paths are not independent, and no proof of the resulting inequality was known. Their paper states five numbered conjectures and four numbered questions of this shape and proves that the weakest of the conjectures implies that critical Bernoulli bond percolation on the hypercubic lattice of dimension at least two has no infinite open cluster.

Conjecture 3 was proved in the Lean development described in Section 10, and Conjecture 1 is a relatively easy consequence of an intermediate inequality of that development, although it is not claimed there. Conjectures 1, 2, 4 and 6 and Questions 5, 7 and 9 are proved here, each in a form stronger than the printed one: Conjecture 2 in its pre-FKG form (1), Conjecture 4 with the vertex realising the minimum prescribed in advance, Conjecture 6 without either of the two hypotheses printed with it, Question 5 with the coefficient vector bounding the probability that the starting vertex reaches both its target and the prescribed set, and Question 9 for every vertex realising its minimum. Question 8 is answered negatively when the vertex realising the minimum is not unique, by the four-vertex example of Section 9, and it is open otherwise.

Everything rests on one inequality. Order the vertices of a finite set  $T$  by the mean of an increasing function  $F$  of the open cluster, and let  $J_x$  be the event that  $x$  is the first vertex of  $T$  that the open cluster of  $o$  reaches. The inequality asserts

$$\sum_{x \in T} \mathbb{P}(J_x) \mathbb{E}F(C_x) \leq \mathbb{E}[F(C_o); o \leftrightarrow T] :$$

the mean of  $F$  at  $o$ , restricted to the event that  $o$  reaches  $T$ , is at least the mean of  $F$  at the vertex through which  $T$  is first reached. For the indicator of the event that  $b$  lies in the cluster this is a statement about connection probabilities, and it is the inequality from which Conjecture 3 was derived.

Two extensions of it carry the present note. The first conditions every expectation on the open cluster missing a prescribed set  $Y$  of vertices, Theorem 4.1. Applied with  $Y$  the single vertex  $a$  at which the mean of  $F$  is smallest, and applied not to  $F$  but to the function  $\Phi_a$  of Section 5, which subtracts from  $F(K)$  the mean of  $F$  at  $a$  in a fresh configuration with all pairs meeting  $K$  deleted, it yields Conjecture 4 with the vertex  $a$  prescribed in advance, Theorem 5.2, and with it Conjectures 1, 2 and 3 and Questions 5 and 7.

The second extension replaces the single vertex  $o$  by a finite set  $R$  of vertices and the cluster  $C_o$  by the union of the clusters of the vertices of  $R$ , Theorem 7.16. For a set of two vertices that union, and the events that it misses a set and that it meets a set, do not depend on the state of the pair joining the two vertices. The inequality therefore holds verbatim in the graph in which that pair is forced open, and Conjecture 6 follows. Question 9 follows from the same theorem with the random set obtained by exposing the pairs at  $o$ .

Sections 2 and 3 fix the model, state the nine numbered items and record the inputs; Section 4 proves Theorem 4.1; Sections 5 to 9 deduce the nine items in the order 4, 2, 1, 3, 7, 5, 6, 9, 8; and Section 10 names the corresponding theorems of the Lean development.

## 2 The model and the nine numbered statements

Let  $V$  be a finite set and let every unordered pair  $e$  of elements of  $V$  be open with probability  $p_e \in [0, 1]$ , independently over the pairs. A pair of weight zero is an absent edge, so this is percolation on a finite weighted graph with vertex set  $V$ . Write  $\mathbb{P}$  and  $\mathbb{E}$  for the resulting probability and expectation. For vertices  $x$  and  $y$ , write  $x \leftrightarrow y$  when there is a path of open pairs from  $x$  to  $y$ , and  $x \nleftrightarrow y$  otherwise; connection is reflexive, so  $x \leftrightarrow x$  always. Write

$$C_x = \{y \in V : x \leftrightarrow y\}$$

for the open cluster of  $x$  and  $\mathcal{C}_x$  for the set of open pairs having at least one endpoint in  $C_x$ . For a set  $K$  of pairs let  $\text{span } K$  be the set of vertices lying in a pair of  $K$ , so that  $C_x = \{x\} \cup \text{span } \mathcal{C}_x$ . For a set  $B$  of vertices put  $C_B = \bigcup_{y \in B} C_y$ , write  $\mathcal{C}_B$  for the set of open pairs having at least one endpoint in  $C_B$ , and write  $\{x \leftrightarrow B\}$  for the event  $\{C_x \cap B \neq \emptyset\}$  and  $\{x \nleftrightarrow B\}$  for its complement. The indicator of an event or of a condition is written **1**. A function  $F$  from subsets of  $V$  to the real numbers is increasing if  $F(K) \leq F(L)$  whenever  $K \subseteq L$ . Every set in this note is finite, and  $A$  is always a nonempty subset of  $V$ .

Kozma and Nitzan write  $\mathbb{P}(0 \leftrightarrow b)$  for what is written  $\mathbb{P}(o \leftrightarrow b)$  here, and they call a function  $f$  of a vertex and a configuration a monotone cluster property if  $f(v, \omega)$  depends only on the open cluster of  $v$  and is increasing in it, and if  $f(v, \omega) = f(w, \omega)$  whenever the pair  $\{v, w\}$  is open. Their five conjectures and four questions are the following. There is no Conjecture 5 and no Question 6.

**Conjecture 2.1** (Kozma and Nitzan, display (1)). Let  $o$  and  $b$  be vertices and let  $A$  be a nonempty set of vertices. Then

$$\mathbb{P}(o \leftrightarrow b) \geq \mathbb{P}(o \leftrightarrow A) \min_{x \in A} \mathbb{P}(x \leftrightarrow b).$$

**Conjecture 2.2** (Kozma and Nitzan, display (2)). Let  $o$  and  $b$  be vertices and let  $A$  be a nonempty set of vertices. Then

$$\mathbb{P}(o \leftrightarrow b) \geq \min_{x \in A} \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b).$$

Kozma and Nitzan call Conjecture 2.1 the post-FKG conjecture and Conjecture 2.2 the pre-FKG

conjecture, and they record the formally stronger form

$$\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) \geq \min_{x \in A} \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b), \quad (1)$$

their display (3).

**Conjecture 2.3** (Kozma and Nitzan, page 15). For every  $\varepsilon > 0$  there is a  $\delta > 0$  such that, for every finite weighted graph, every nonempty set  $A$  of vertices and all vertices  $o$  and  $b$  with  $\mathbb{P}(o \leftrightarrow A) > 1 - \delta$  and with  $\mathbb{P}(x \leftrightarrow b) > 1 - \delta$  for every  $x \in A$ , one has  $\mathbb{P}(o \leftrightarrow b) > 1 - \varepsilon$ .

**Conjecture 2.4** (Kozma and Nitzan, page 32). Let  $f$  be a monotone cluster property, let  $A$  be a nonempty set of vertices and let  $o$  be a vertex. Then

$$\mathbb{E}[f(o, \cdot); o \leftrightarrow A] \geq \min_{x \in A} \mathbb{E}[f(x, \cdot); o \leftrightarrow A].$$

For a pair  $e$ , write  $\mathbb{P}^e$  and  $\mathbb{E}^e$  for the probability and the expectation in the percolation measure in which the weight of  $e$  is replaced by one and all other weights are unchanged. Kozma and Nitzan write  $G \downarrow e$  for the corresponding graph, which is the contraction of  $e$  as far as connection events are concerned.

**Conjecture 2.5** (Kozma and Nitzan, displays (39) and (40)). Let  $A$  be a nonempty set of vertices, let  $b$  be a vertex, let  $a \in A$  minimise  $\mathbb{P}(x \leftrightarrow b)$  over  $x \in A$ , and let  $e = \{v, w\}$  be a pair. Assume

$$\mathbb{P}(x \leftrightarrow b) \geq \mathbb{P}(x \leftrightarrow A) \mathbb{P}(a \leftrightarrow b) \quad \text{for } x = v \text{ and } x = w. \quad (2)$$

Then

$$\mathbb{P}^e(v \leftrightarrow b) \geq \mathbb{P}^e(v \leftrightarrow A) \mathbb{P}^e(a \leftrightarrow b). \quad (3)$$

**Question 2.6** (Kozma and Nitzan, display (38)). Let  $A$  be a nonempty set of vertices and let  $o$  be a vertex. Are there numbers  $c_x \geq 0$  for  $x \in A$  with  $\sum_{x \in A} c_x = 1$ , not depending on  $b$ , such that

$$\mathbb{P}(o \leftrightarrow b) \geq \sum_{x \in A} c_x \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b) \quad \text{for every vertex } b?$$

The last three questions all ask whether a specified vertex  $a \in A$  satisfies the inequality of the paper's display (41),

$$\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) \geq \mathbb{P}(o \leftrightarrow A, a \leftrightarrow b). \quad (4)$$

**Question 2.7** (Kozma and Nitzan, page 36). Does (4) hold when  $a$  minimises  $\mathbb{P}(x \leftrightarrow b)$  over  $x \in A$ ?

**Question 2.8** (Kozma and Nitzan, page 36). Does (4) hold when  $a$  minimises  $\mathbb{P}(x \leftrightarrow b, o \leftrightarrow A)$  over  $x \in A$ ?

**Question 2.9** (Kozma and Nitzan, page 36). Write  $\mathbb{P}_\star$  and  $\mathbb{E}_\star$  for the probability and the expectation in the percolation measure in which every pair incident to  $o$  is closed and all other weights are unchanged. Does (4) hold, with all probabilities evaluated in the original measure, when  $a$  minimises  $\mathbb{P}_\star(x \leftrightarrow b)$  over  $x \in A$ ?

The minimand in Question 2.8 is the probability of the joint event that  $x$  reaches  $b$  and that  $o$  does not reach  $A$ ; it is not the minimand of (4) itself, which is treated by (1).

### 3 The two inputs

Two families of inequalities are taken as given. The first is the conditional positive association of van den Berg, Häggström and Kahn.

**Theorem 3.1** (van den Berg, Häggström and Kahn). *Let  $s$  be a vertex, let  $X_1$  and  $X_2$  be sets of vertices not containing  $s$  and let  $g_1$  and  $g_2$  be nonnegative increasing functions of sets of pairs. Then*

$$\mathbb{E}[g_1(\mathcal{C}_s); s \leftrightarrow X_1] \mathbb{E}[g_2(\mathcal{C}_s); s \leftrightarrow X_2] \leq \mathbb{E}[g_1(\mathcal{C}_s)g_2(\mathcal{C}_s); s \leftrightarrow X_1 \cap X_2] \mathbb{P}(s \leftrightarrow X_1 \cup X_2).$$

Moreover, let  $S_1$  and  $S_2$  be sets of vertices and put  $D = \{S_1 \leftrightarrow S_2\}$ . If  $g$  and  $h$  are real functions of two sets of pairs,  $g$  increasing in the first argument and decreasing in the second and  $h$  decreasing in the first and increasing in the second, then

$$\mathbb{P}(D) \mathbb{E}[g(\mathcal{C}_{S_1}, \mathcal{C}_{S_2}) h(\mathcal{C}_{S_1}, \mathcal{C}_{S_2}); D] \leq \mathbb{E}[g(\mathcal{C}_{S_1}, \mathcal{C}_{S_2}); D] \mathbb{E}[h(\mathcal{C}_{S_1}, \mathcal{C}_{S_2}); D].$$

Taking  $X_1 = X_2$  in the first display gives the conditional positive association of two increasing functions of a single cluster,

$$\mathbb{E}[g_1(\mathcal{C}_s); s \leftrightarrow X] \mathbb{E}[g_2(\mathcal{C}_s); s \leftrightarrow X] \leq \mathbb{P}(s \leftrightarrow X) \mathbb{E}[g_1(\mathcal{C}_s)g_2(\mathcal{C}_s); s \leftrightarrow X]. \quad (5)$$

Both sides of (5) increase by the same amount when a constant is added to  $g_1$  or to  $g_2$ , and each takes finitely many values, so the nonnegativity assumption may be dropped in (5): monotonicity of  $g_1$  and  $g_2$  suffices. Taking  $g_1 = g_2 = 1$  in the first display gives the Harris inequality for the two decreasing events  $\{s \leftrightarrow X_1\}$  and  $\{s \leftrightarrow X_2\}$ . The Harris inequality for two increasing events is used repeatedly and without further comment. The Ahlswede and Daykin four functions theorem is used once, in Lemma 7.11.

The second input is a family of covariance inequalities for one open cluster, conditioned on that cluster missing a prescribed set of vertices, and discounted by a finite list of auxiliary vertices. Fix a vertex  $s$ , a set  $Y$  of vertices with  $s \notin Y$ , a list  $z_1, \dots, z_\ell$  of distinct vertices and two further distinct vertices  $o$  and  $v$ , with all the named vertices outside  $Y$  and pairwise distinct. Write  $\nu$  for the conditional law  $\mathbb{P}(\cdot \mid s \leftrightarrow Y)$  and put

$$c_j(u) = \mathbb{P}(u \in C_{z_j} \mid z_j \leftrightarrow Y_j), \quad \tau = \mathbb{P}(o \in C_v \mid v \leftrightarrow Y_{\ell+1}), \quad (6)$$

where  $Y_j = \{s\} \cup Y \cup \{z_1, \dots, z_{j-1}\}$  for  $1 \leq j \leq \ell + 1$ . For a real function  $\varphi$  on  $V$  put  $\varphi^0 = \varphi$  and

$$\varphi^j(u) = \varphi^{j-1}(u) - c_j(u) \varphi^{j-1}(z_j) \quad (1 \leq j \leq \ell), \quad \mathbb{M}[\varphi] = \varphi^\ell(o) - \tau \varphi^\ell(v). \quad (7)$$

The  $j$ th step removes from the value at every vertex the part explained by  $z_j$ , at the rate at which  $z_j$  reaches that vertex while missing everything named before it. All the constants are computed under  $\mathbb{P}$  and not under  $\nu$ .

**Theorem 3.2.** *Let  $s$ ,  $Y$ , the list  $z_1, \dots, z_\ell$  and the vertices  $o$  and  $v$  be as above, let every pair weight lie strictly between zero and one, and let  $g$  be an increasing function of sets of pairs. Then*

$$\mathbb{M}[u \mapsto \text{Cov}_\nu(g(\mathcal{C}_s), \mathbf{1}\{u \in C_s\})] \geq 0.$$

For  $\ell = 0$  the conclusion reads

$$\text{Cov}_\nu(g(\mathcal{C}_s), \mathbf{1}\{o \in C_s\}) \geq \mathbb{P}(o \in C_v \mid v \leftrightarrow \{s\} \cup Y) \text{Cov}_\nu(g(\mathcal{C}_s), \mathbf{1}\{v \in C_s\}),$$

and if in addition  $Y = \emptyset$  and  $g = \mathbf{1}\{v \in \cdot\}$  it is the Harris inequality.

One further construction is recorded here, since it is a property of the objects appearing in the proof of Theorem 3.2 rather than of its statement, and Section 7 uses it. Condition on the set  $\mathcal{C}_Y$  of open pairs met by paths from  $Y$ . On the event  $D = \{s \leftrightarrow Y\}$ , the vertex set  $C_Y = Y \cup \text{span } \mathcal{C}_Y$  misses  $C_s$ , every pair with exactly one endpoint in  $C_Y$  is closed, and the pairs with both endpoints outside  $C_Y$  retain their original independent weights. Consequently, writing  $\bar{g}$  for the mean of  $g(\mathcal{C}_s)$  in a fresh configuration in which every pair meeting  $C_Y$  is deleted, the random variable

$$\Theta = \mathbf{1}_D(g(\mathcal{C}_s) - \bar{g}) \quad (8)$$

has  $\mathbb{E}[\Theta \mid \mathcal{C}_Y] = 0$ . For a set  $K$  of vertices disjoint from  $\{s\} \cup Y$ , let  $\Psi(K)$  be minus the mean of  $\Theta$  computed in the configuration in which every pair meeting  $K$  is deleted. Then  $\Psi(\emptyset) = 0$ , the function  $\Psi$  is nonnegative and increasing, and for every vertex  $z$  and every set  $B$  containing  $\{s\} \cup Y$  but not  $z$ ,

$$\mathbb{E}[\Theta; z \leftrightarrow B] = -\mathbb{E}[\Psi(C_z); z \leftrightarrow B]. \quad (9)$$

Both the vanishing of the conditional mean of  $\Theta$  and the identity (9) express the domain Markov property of percolation: given the complete set of open pairs met by paths from a set, the configuration outside the cluster of that set is fresh.

Two properties of the conditioned covariance itself are recorded for the same reason. For a set  $W$  of vertices write  $\mathbb{P}_W$  and  $\mathbb{E}_W$  for percolation in which every pair with an endpoint outside  $W$  is closed and every pair with both endpoints in  $W$  keeps its weight, and write  $\text{Cov}_W$  for the covariance under  $\mathbb{P}_W$ . For a set  $X$  of vertices containing  $s$  and a vertex  $v \notin X$ , put

$$\eta_v(W) = \text{Cov}_W(g(\mathcal{C}_s), \mathbf{1}\{v \leftrightarrow X\}) \quad \text{for } v \in W, \quad \eta_v(W) = 0 \quad \text{for } v \notin W. \quad (10)$$

The construction underlying Theorem 3.2 yields, for every  $W$  and every  $Q \subseteq W$ ,

$$\mathbb{E}_W[\mathbf{1}\{s \leftrightarrow Q\} \eta_v(W \setminus C_Q)] \mathbb{P}_W(v \leftrightarrow X) \leq \mathbb{P}_W(v \leftrightarrow X \cup Q) \eta_v(W), \quad (11)$$

and the left side of (11) decreases as  $Q$  grows.

We turn to the inequality from which all of the numbered statements are deduced.

## 4 A lower bound conditioned on missing a fixed set

Fix a set  $Y$  of vertices and an increasing function  $F$  of vertex sets. For a vertex  $x$  with  $\mathbb{P}(x \leftrightarrow Y) > 0$  put

$$m_x = \frac{\mathbb{E}[F(C_x); x \leftrightarrow Y]}{\mathbb{P}(x \leftrightarrow Y)}, \quad (12)$$

the mean of  $F$  at  $x$  given that  $x$  misses  $Y$ . Let  $T$  be a finite set of vertices disjoint from  $Y$  with  $\mathbb{P}(x \leftrightarrow Y) > 0$  for every  $x \in T$ , and let  $r$  be an injective real function on  $T$  such that  $m_x \leq m_y$  whenever  $r(x) < r(y)$ . Such a function exists: order  $T$  by the values  $m_x$  and break ties arbitrarily. For  $x \in T$  and a vertex  $o$  write

$$J_x = \{o \leftrightarrow Y\} \cap \{o \leftrightarrow x\} \cap \bigcap_{y \in T, r(y) < r(x)} \{o \leftrightarrow y\} \quad (13)$$

for the event that  $o$  misses  $Y$  and that  $x$  is the vertex of smallest value of  $r$  that the open cluster of  $o$  reaches in  $T$ . The events  $J_x$  for  $x \in T$  are disjoint and their union is  $\{o \leftrightarrow Y\} \cap \{o \leftrightarrow T\}$ . Put

$$\Lambda_o(T) = \mathbb{E}[F(C_o); o \leftrightarrow Y, o \leftrightarrow T] - \sum_{x \in T} \mathbb{P}(J_x) m_x. \quad (14)$$

On  $J_x$  the vertex  $x$  has the smallest value of  $r$  among the vertices of  $T$  in  $C_o$ , hence, by the assumption on  $r$ , the smallest value of  $m$ , and on  $J_x$  the clusters of  $o$  and  $x$  coincide. Consequently (14) admits the expression

$$\Lambda_o(T) = \mathbb{E}\left[\mathbf{1}\{o \leftrightarrow Y, o \leftrightarrow T\} \left(F(C_o) - \min_{x \in T \cap C_o} m_x\right)\right], \quad (15)$$

which does not refer to  $r$ . Two positions of  $o$  are settled by (15). If  $o \in Y$  then the event  $\{o \leftrightarrow Y\}$  is empty and  $\Lambda_o(T) = 0$ . If  $o \in T$  then  $o \in T \cap C_o$  on  $\{o \leftrightarrow Y\}$ , so the minimum in (15) is at most  $m_o$ , whence  $\Lambda_o(T) \geq \mathbb{E}[F(C_o) - m_o; o \leftrightarrow Y] = 0$  by the definition (12) of  $m_o$ .

**Theorem 4.1.** *Let  $V$  be a finite vertex set with pair weights in  $[0, 1]$ , let  $Y \subseteq V$ , let  $F$  be an increasing function of subsets of  $V$ , let  $T \subseteq V \setminus Y$  satisfy  $\mathbb{P}(x \leftrightarrow Y) > 0$  for every  $x \in T$ , let  $r$  be injective on  $T$  with  $m_x \leq m_y$  whenever  $r(x) < r(y)$ , and let  $o \in V$ . Then  $\Lambda_o(T) \geq 0$ , that is*

$$\sum_{x \in T} \mathbb{P}(J_x) m_x \leq \mathbb{E}[F(C_o); o \leftrightarrow Y, o \leftrightarrow T].$$

For  $Y = \emptyset$  this is the inequality quoted in the introduction. No sign hypothesis on  $F$  is needed. The proof removes from  $T$  the vertex at which  $m$  is largest and transfers the resulting deficit from that vertex back to  $o$ . The transfer step is the following inequality, which is what Theorem 3.2 supplies.

**Proposition 4.2.** *Let  $V$  be a finite vertex set with pair weights in  $[0, 1]$ , let  $Y \subseteq V$ , let  $F$  be an increasing function of subsets of  $V$ , let  $T \subseteq V \setminus Y$  satisfy  $\mathbb{P}(x \leftrightarrow Y) > 0$  for every  $x \in T$ , let  $r$  be injective on  $T$  with  $m_x \leq m_y$  whenever  $r(x) < r(y)$ , and let  $v \in V$  lie outside  $Y \cup T$ . Then, for every vertex  $o$ ,*

$$\mathbb{P}(v \leftrightarrow Y \cup T, o \leftrightarrow v) \Lambda_v(T) \leq \mathbb{P}(v \leftrightarrow Y \cup T) \Lambda_o(T).$$

*Proof.* Assume first that every pair weight lies strictly between zero and one and that  $o$  lies outside  $Y \cup T$  and differs from  $v$ ; the remaining positions of  $o$  are treated at the end. Let  $z_1, \dots, z_\ell$  be distinct vertices outside  $Y \cup T \cup \{o, v\}$  and let  $M$  be the functional (7) built from the constants (6) with  $Y$  replaced by  $Y \cup T$  at every step, so that  $z_j$  is conditioned to miss  $Y \cup T \cup \{z_1, \dots, z_{j-1}\}$  and the final constant is  $\tau = \mathbb{P}(o \in C_v \mid v \leftrightarrow Y \cup T \cup \{z_1, \dots, z_\ell\})$ . We show by induction on the cardinality of  $T$ , simultaneously for every such list and every pair of vertices  $o$  and  $v$ , that

$$M[u \mapsto \Lambda_u(T)] \geq 0. \quad (16)$$

For  $T = \emptyset$  each  $\Lambda_u(\emptyset)$  vanishes. Let  $T$  be nonempty, let  $k \in T$  have the largest value of  $r$  and put  $T_0 = T \setminus \{k\}$ . Write  $E = \{k \leftrightarrow Y \cup T_0\}$ .

We first record an identity. Since  $k$  has the largest value of  $r$ , adding  $k$  to  $T_0$  leaves the events  $J_x$  for  $x \in T_0$  unchanged and creates the single new event  $\{u \leftrightarrow Y\} \cap \{u \leftrightarrow k\} \cap \{u \leftrightarrow T_0\}$ , which equals  $E \cap \{u \leftrightarrow k\}$ : indeed on  $\{u \leftrightarrow k\}$  the clusters of  $u$  and  $k$  coincide. On that event  $F(C_u) = F(C_k)$ , so

$$\Lambda_u(T) = \Lambda_u(T_0) + \mathbb{E}[F(C_k); E \cap \{u \leftrightarrow k\}] - \mathbb{P}(E \cap \{u \leftrightarrow k\}) m_k. \quad (17)$$

The last two terms on the right are handled together. Write  $\text{Cov}^*(u)$  for  $\mathbb{P}(E)\mathbb{E}[F(C_k); E \cap \{u \leftrightarrow k\}] - \mathbb{E}[F(C_k); E]\mathbb{P}(E \cap \{u \leftrightarrow k\})$ , the conditioned covariance of Theorem 3.2 written without denominators, for the vertex  $k$ , the set  $Y \cup T_0$  and the increasing function  $\mathcal{C} \mapsto F(\{k\} \cup \text{span } \mathcal{C})$ , and put  $\gamma_k = m_k \mathbb{P}(E) - \mathbb{E}[F(C_k); E]$ . Expanding both sides gives

$$\mathbb{P}(E) \left( \mathbb{E}[F(C_k); E \cap \{u \leftrightarrow k\}] - \mathbb{P}(E \cap \{u \leftrightarrow k\}) m_k \right) = \text{Cov}^*(u) - \gamma_k \mathbb{P}(E \cap \{u \leftrightarrow k\}). \quad (18)$$

Theorem 3.2 for the vertex  $k$ , the set  $Y \cup T_0$ , the same list and the same two vertices gives  $M[\text{Cov}^*] \geq 0$ . Theorem 3.2 for the same vertex and set and for the increasing function  $\mathcal{C} \mapsto \mathbf{1}\{\mathcal{C} \neq \emptyset\}$  gives  $M[u \mapsto \mathbb{P}(E \cap \{u \leftrightarrow k\})] \geq 0$ . Indeed  $M$  evaluates its argument only at  $o$ , at  $v$  and at the vertices of the list, all of which differ from  $k$ ; on  $\{u \leftrightarrow k\}$  with  $u \neq k$  the cluster of  $k$  contains an open pair, and on  $E$  the mean of that indicator is  $\mathbb{P}(E)$  minus the probability that  $k$  misses  $Y \cup T_0$  and lies in no open pair, so the covariance in question is the latter probability times  $\mathbb{P}(E \cap \{u \leftrightarrow k\})$ , and that probability is positive because all weights are strictly below one. Finally,

$$\gamma_k \leq \Lambda_k(T_0). \quad (19)$$

Indeed the definition of  $m_k$  gives  $\mathbb{E}[(F(C_k) - m_k); k \leftrightarrow Y] = 0$ ; splitting  $\{k \leftrightarrow Y\}$  into  $E$  and  $\{k \leftrightarrow Y\} \cap \{k \leftrightarrow T_0\}$  turns this into  $\gamma_k = \mathbb{E}[(F(C_k) - m_k); k \leftrightarrow Y, k \leftrightarrow T_0]$ , and expanding the latter over the disjoint events  $J_x$  of (13) taken with  $k$  in place of  $o$  gives  $\gamma_k = \Lambda_k(T_0) + \sum_{x \in T_0} \mathbb{P}(J_x)(m_x - m_k)$ , in which every summand is at most zero because  $r(x) < r(k)$ . Now apply  $M$  to (17), multiply by  $\mathbb{P}(E) > 0$  and use (18):

$$\mathbb{P}(E) M[\Lambda_u(T)] = \mathbb{P}(E) M[\Lambda_u(T_0)] + M[\text{Cov}^*] - \gamma_k M[u \mapsto \mathbb{P}(E \cap \{u \leftrightarrow k\})].$$

The middle term is nonnegative. Replacing  $\gamma_k$  by the larger  $\Lambda_k(T_0)$ , which is legitimate because the last margin is nonnegative, and writing  $\mathbb{P}(E \cap \{u \leftrightarrow k\}) = \mathbb{P}(E) \mathbb{P}(u \in C_k \mid k \leftrightarrow Y \cup T_0)$ , the right side is at least  $\mathbb{P}(E)$  times the margin of  $u \mapsto \Lambda_u(T_0)$  for the list obtained by inserting  $k$  in front of  $z_1, \dots, z_\ell$ , by the recursion in (7) and linearity; that margin is nonnegative by the induction hypothesis. Dividing by  $\mathbb{P}(E)$  gives (16).

Take now  $\ell = 0$  in (16). The conclusion is  $\Lambda_o(T) \geq \tau \Lambda_v(T)$  with  $\tau = \mathbb{P}(o \in C_v \mid v \leftrightarrow Y \cup T)$ , which is the assertion after multiplication by  $\mathbb{P}(v \leftrightarrow Y \cup T) > 0$ . If  $o = v$  the assertion is trivial. If  $o$  lies in  $Y$

or in  $T$  then the event  $\{v \leftrightarrow Y \cup T\} \cap \{o \leftrightarrow v\}$  is empty, so the left side vanishes, while the right side is nonnegative by the two cases settled after (15).

It remains to pass from weights strictly inside the unit interval to weights in  $[0, 1]$ . The right side of (15) is a finite sum, over the configurations, of the probability of the configuration, a polynomial in the pair weights, times a value depending on the weights only through the finitely many numbers  $m_x$ , each a ratio of polynomials whose denominator is positive at the given weight vector. Both sides of the asserted inequality are therefore continuous at that vector. Approximate it by weight vectors with all coordinates strictly inside the unit interval and choose for each of them an injective  $r$  ordered by its own numbers  $m_x$ ; by (15) neither side depends on that choice, so the case already proved applies to each of them, and continuity gives the inequality at the given vector.  $\square$

*Proof of Theorem 4.1.* Induct on the cardinality of  $T$ . For  $T = \emptyset$  the quantity (14) vanishes. Let  $T$  be nonempty, let  $k \in T$  have the largest value of  $r$ , put  $T_0 = T \setminus \{k\}$  and  $E = \{k \leftrightarrow Y \cup T_0\}$ , and abbreviate

$$P_1 = \mathbb{P}(E), \quad P_2 = \mathbb{P}(E \cap \{o \leftrightarrow k\}), \quad I_1 = \mathbb{E}[F(C_k); E], \quad I_2 = \mathbb{E}[F(C_k); E \cap \{o \leftrightarrow k\}].$$

The identity (17) with  $u = o$  reads  $\Lambda_o(T) = \Lambda_o(T_0) + I_2 - m_k P_2$ . Apply (5) with  $s = k$ , with  $X = Y \cup T_0$  and with the two increasing functions  $\mathcal{C} \mapsto F(\{k\} \cup \text{span } \mathcal{C})$  and the indicator of the event that  $o$  lies in the cluster of  $k$ ; the result is  $P_2 I_1 \leq P_1 I_2$ , which rearranges to

$$P_1(m_k P_2 - I_2) \leq P_2(m_k P_1 - I_1) = P_2 \gamma_k.$$

By (19) and  $P_2 \geq 0$  the right side is at most  $P_2 \Lambda_k(T_0)$ , and by Proposition 4.2 with  $v = k$  that is at most  $P_1 \Lambda_o(T_0)$ . Hence

$$P_1(m_k P_2 - I_2) \leq P_1 \Lambda_o(T_0), \quad \text{that is} \quad P_1 \Lambda_o(T) \geq 0.$$

If  $P_1 > 0$  this is the assertion. If  $P_1 = 0$  then  $P_2 = 0$  and  $I_2 = 0$ , so  $\Lambda_o(T) = \Lambda_o(T_0)$ , which is nonnegative by the induction hypothesis.  $\square$

The next section turns Theorem 4.1 into a statement in which the vertex realising the minimum is fixed before the event  $\{o \leftrightarrow A\}$  is imposed.

## 5 A prescribed vertex works in Conjecture 4

The obstacle to reading Conjecture 2.4 off Theorem 4.1 is that the numbers  $m_x$  of (12) are conditional means, whereas Conjecture 2.4 compares unconditional ones. The following function removes the difference. For a vertex  $a$ , an increasing  $F$  and a set  $K$  of vertices, write  $\Phi_a(K)$  for  $F(K)$  minus the mean of  $F(C_a)$  in a fresh configuration in which every pair meeting  $K$  is closed.

**Lemma 5.1.** *Let  $a$  and  $o$  be vertices and let  $F$  be increasing. Then  $\Phi_a$  is increasing, and:*

(i) *for every family  $\mathcal{S}$  of sets of pairs,*

$$\mathbb{E}[F(C_o) - F(C_a); o \leftrightarrow a, \mathcal{C}_o \in \mathcal{S}] = \mathbb{E}[\Phi_a(C_o); o \leftrightarrow a, \mathcal{C}_o \in \mathcal{S}];$$

(ii)  $\mathbb{E}[\Phi_a(C_o); o \leftrightarrow a] = \mathbb{E}F(C_o) - \mathbb{E}F(C_a)$ .

*Proof.* Enlarging  $K$  increases  $F(K)$  and deletes more pairs, which decreases the subtracted mean, so  $\Phi_a$  is increasing.

For (i), condition on  $\mathcal{C}_o$ . On  $\{o \leftrightarrow a\}$  the vertex  $a$  lies outside  $C_o$ , every pair with exactly one endpoint in  $C_o$  is closed, and the pairs with both endpoints outside  $C_o$  retain their original weights. Hence the conditional mean of  $F(C_a)$  given  $\mathcal{C}_o$  is  $F(C_o) - \Phi_a(C_o)$ . Both events in the statement are determined by  $\mathcal{C}_o$ , so summing over the values of  $\mathcal{C}_o$  in  $\mathcal{S}$  gives (i).

For (ii), take  $\mathcal{S}$  to consist of all sets of pairs in (i). On  $\{o \leftrightarrow a\}$  the two clusters coincide, so  $F(C_o) - F(C_a)$  vanishes there and the left side of (i) equals  $\mathbb{E}[F(C_o) - F(C_a)]$ .  $\square$

**Theorem 5.2.** *Let  $A$  be a nonempty finite set of vertices, let  $a \in A$ , let  $o$  be a vertex and let  $F$  be increasing with  $\mathbb{E}F(C_a) \leq \mathbb{E}F(C_x)$  for every  $x \in A$ . Then*

$$\mathbb{E}[F(C_a); o \leftrightarrow A] \leq \mathbb{E}[F(C_o); o \leftrightarrow A].$$

*Proof.* Let  $T$  be the set of  $x \in A \setminus \{a\}$  with  $\mathbb{P}(x \leftrightarrow a) > 0$ . Apply Theorem 4.1 with  $Y = \{a\}$ , with the set  $T$ , with the increasing function  $\Phi_a$  of Lemma 5.1 and with the vertex  $o$ . By Lemma 5.1(ii) applied at  $x$  in place of  $o$ , the number (12) attached to  $x \in T$  is

$$\frac{\mathbb{E}[\Phi_a(C_x); x \leftrightarrow a]}{\mathbb{P}(x \leftrightarrow a)} = \frac{\mathbb{E}F(C_x) - \mathbb{E}F(C_a)}{\mathbb{P}(x \leftrightarrow a)} \geq 0,$$

so the sum subtracted in (14) is nonnegative and Theorem 4.1 gives

$$\mathbb{E}[\Phi_a(C_o); o \leftrightarrow a, o \leftrightarrow T] \geq 0.$$

The event  $\{o \leftrightarrow T\}$  is determined by  $C_o$ , so Lemma 5.1(i) converts this into

$$\mathbb{E}[F(C_o) - F(C_a); o \leftrightarrow a, o \leftrightarrow T] \geq 0. \quad (20)$$

Split  $\{o \leftrightarrow A\}$  into  $\{o \leftrightarrow a\}$ , on which the integrand  $F(C_o) - F(C_a)$  vanishes, and  $\{o \leftrightarrow a\} \cap \{o \leftrightarrow A \setminus \{a\}\}$ . The latter differs from  $\{o \leftrightarrow a\} \cap \{o \leftrightarrow T\}$  only by configurations in which  $o$  misses  $a$  and reaches some  $x \in A \setminus \{a\}$  outside  $T$ ; such a configuration lies in the null event  $\{x \leftrightarrow a\}$ , and a finite union of null events is null. Hence the left side of (20) equals  $\mathbb{E}[F(C_o) - F(C_a); o \leftrightarrow A]$ .  $\square$

Conjecture 2.4 follows at once by choosing  $a$  to realise the minimum of  $\mathbb{E}F(C_x)$  over  $x \in A$ , provided every monotone cluster property is of the form  $\omega \mapsto F(C_v(\omega))$  for an increasing  $F$ . That is the next lemma, whose hypotheses omit the requirement that  $f(v, \omega)$  depend only on  $C_v$ , since that follows from the requirement that  $f(v, \omega)$  be increasing in  $C_v$ .

**Lemma 5.3.** *Let  $f$  be a function of a vertex and a configuration such that  $f(v, \omega) \leq f(v, \xi)$  whenever  $C_v(\omega) \subseteq C_v(\xi)$  and such that  $f(v, \omega) = f(w, \omega)$  whenever the pair  $\{v, w\}$  is open in  $\omega$ . Then there is an increasing function  $F$  of vertex sets with  $f(v, \omega) = F(C_v(\omega))$  for every vertex  $v$  and every configuration  $\omega$ .*

*Proof.* For a nonempty vertex set  $S$  let  $\omega_S$  be the configuration in which the open pairs are exactly those with both endpoints in  $S$ . Every vertex of  $S$  has open cluster  $S$  in  $\omega_S$ , and for distinct  $v, w \in S$  the pair  $\{v, w\}$  is open in  $\omega_S$ , so  $f(v, \omega_S)$  does not depend on the choice of  $v \in S$ ; call the common value  $F(S)$ . If  $S \subseteq S_1$  are nonempty and  $v \in S$  then  $C_v(\omega_S) = S \subseteq S_1 = C_v(\omega_{S_1})$ , so  $F(S) \leq F(S_1)$ . Extend  $F$  to the empty set by a value at most  $F(S)$  for every nonempty  $S$ . Finally, for a vertex  $v$  and a configuration  $\omega$  put  $S = C_v(\omega)$ ; then  $C_v(\omega_S) = S = C_v(\omega)$ , and the hypothesis applied in both directions gives  $f(v, \omega) = f(v, \omega_S) = F(S)$ .  $\square$

**Corollary 5.4.** *Conjecture 2.4 holds. In particular it holds for  $f(v, \omega) = |C_v|$  and for  $f(v, \omega) = \mathbf{1}\{|C_v| \geq k\}$  for every  $k$ .*

*Proof.* By Lemma 5.3 write  $f(v, \cdot) = F(C_v)$  with  $F$  increasing, choose  $a \in A$  realising the minimum of  $\mathbb{E}F(C_x)$  over  $x \in A$ , and apply Theorem 5.2. The two displayed examples are increasing functions of the vertex set.  $\square$

The remaining conjectures of the paper follow by specialising  $F$  to an indicator.

**Corollary 5.5.** *The inequality (1) holds, and so do Conjectures 2.1, 2.2 and 2.3; Conjecture 2.3 holds with  $\delta = \varepsilon/2$ . Question 2.7 has an affirmative answer.*



*Proof.* Take  $F = \mathbf{1}\{b \in \cdot\}$ , so that  $\mathbb{E}[F(C_x); o \leftrightarrow A] = \mathbb{P}(x \leftrightarrow b, o \leftrightarrow A)$  and  $\mathbb{E}F(C_x) = \mathbb{P}(x \leftrightarrow b)$ . Corollary 5.4 is then (1), and dropping the event  $\{o \leftrightarrow A\}$  on the left gives Conjecture 2.2. If  $a$  minimises  $\mathbb{P}(x \leftrightarrow b)$  over  $x \in A$  then Theorem 5.2 gives  $\mathbb{P}(a \leftrightarrow b, o \leftrightarrow A) \leq \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$ , which is (4), so Question 2.7 has an affirmative answer.

For Conjecture 2.1, the events  $\{x \leftrightarrow b\}$  and  $\{o \leftrightarrow A\}$  are both increasing, so the Harris inequality gives  $\mathbb{P}(o \leftrightarrow A)\mathbb{P}(x \leftrightarrow b) \leq \mathbb{P}(x \leftrightarrow b, o \leftrightarrow A)$  for every  $x \in A$ . Taking the minimum over  $x \in A$  and using (1),

$$\mathbb{P}(o \leftrightarrow A) \min_{x \in A} \mathbb{P}(x \leftrightarrow b) \leq \min_{x \in A} \mathbb{P}(x \leftrightarrow b, o \leftrightarrow A) \leq \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) \leq \mathbb{P}(o \leftrightarrow b).$$

For Conjecture 2.3 take  $\delta = \varepsilon/2$ . If  $\varepsilon \geq 2$  then  $1 - \varepsilon \leq -1 < \mathbb{P}(o \leftrightarrow b)$  and there is nothing to prove. Otherwise  $1 - \varepsilon/2$  is positive, and Conjecture 2.1 gives  $\mathbb{P}(o \leftrightarrow b) > (1 - \varepsilon/2)^2 = 1 - \varepsilon + \varepsilon^2/4 \geq 1 - \varepsilon$ .  $\square$

## 6 One coefficient vector for every target

Question 2.6 asks for one probability vector on  $A$  that works simultaneously for every vertex  $b$ . Corollary 5.4 produces, for each  $b$ , a vertex of  $A$  that works for that  $b$ ; the passage from one to the other is finite-dimensional convexity. The first step is to apply Corollary 5.4 not to a single indicator but to a nonnegative weighted count of the vertices of the cluster.

**Lemma 6.1.** *Let  $A$  be a nonempty finite set of vertices, let  $o$  be a vertex and let  $y_b \geq 0$  for  $b \in V$ . Then there is a vertex  $a \in A$  with*

$$\sum_{b \in V} y_b \mathbb{P}(o \leftrightarrow A, a \leftrightarrow b) \leq \sum_{b \in V} y_b \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A).$$

*Proof.* Put  $F(K) = \sum_{b \in V} y_b \mathbf{1}\{b \in K\}$ , an increasing function of vertex sets because each  $y_b$  is nonnegative. For every vertex  $x$ ,

$$\mathbb{E}[F(C_x); o \leftrightarrow A] = \sum_{b \in V} y_b \mathbb{P}(x \leftrightarrow b, o \leftrightarrow A).$$

Let  $a \in A$  realise the minimum of  $\mathbb{E}[F(C_x); o \leftrightarrow A]$  over  $x \in A$ . By Corollary 5.4, the left side of the assertion is  $\mathbb{E}[F(C_a); o \leftrightarrow A]$ , which is at most  $\mathbb{E}[F(C_o); o \leftrightarrow A]$ , and the latter is the right side.  $\square$

**Theorem 6.2.** *For every finite weighted graph, every nonempty finite  $A \subseteq V$  and every vertex  $o$  there are numbers  $c_x \geq 0$  for  $x \in A$  with  $\sum_{x \in A} c_x = 1$  such that*

$$\sum_{x \in A} c_x \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b) \leq \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) \leq \mathbb{P}(o \leftrightarrow b) \quad \text{for every vertex } b.$$

*In particular Question 2.6 has an affirmative answer.*

*Proof.* The second inequality of the assertion is event inclusion, so only the first has to be proved. Regard the numbers  $M(x, b) = \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b)$  for  $x \in A$ , and the numbers  $\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$  assembled into a vector  $\mathbf{t}$ , as vectors indexed by  $b \in V$ . Let  $\mathcal{P}$  be the convex hull of the finitely many vectors  $M(x, \cdot)$  for  $x \in A$ , a compact convex subset of the space of real vectors indexed by  $V$ , and let  $\mathcal{D}$  be the set of differences of a point of  $\mathcal{P}$  and a vector with nonnegative entries. The set  $\mathcal{D}$  is convex, and it is closed because it is the sum of a compact set and a closed set.

Suppose  $\mathbf{t} \notin \mathcal{D}$ . Strict separation of the point  $\mathbf{t}$  from the closed convex set  $\mathcal{D}$  gives a vector  $\mathbf{y}$  indexed by  $V$  and a real number  $\alpha$  with  $\langle \mathbf{y}, \mathbf{t} \rangle < \alpha \leq \langle \mathbf{y}, \mathbf{z} \rangle$  for every  $\mathbf{z} \in \mathcal{D}$ . Fix  $b \in V$ , let  $\mathbf{e}_b$  be the vector with entry one at  $b$  and zero elsewhere, and fix a point  $\mathbf{z}$  of the nonempty set  $\mathcal{D}$ . Then  $\mathbf{z} - n\mathbf{e}_b$  lies in  $\mathcal{D}$  for every positive integer  $n$ , so  $\langle \mathbf{y}, \mathbf{z} \rangle - ny_b \geq \alpha$  for every such  $n$ , which forces  $y_b \geq 0$ . Applying the separating inequality at the points  $M(x, \cdot)$  of  $\mathcal{P} \subseteq \mathcal{D}$  gives  $\sum_b y_b M(x, b) \geq \alpha > \sum_b y_b \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$  for every  $x \in A$ , contradicting Lemma 6.1. Hence  $\mathbf{t} \in \mathcal{D}$ : there are  $c_x \geq 0$  with  $\sum_{x \in A} c_x = 1$  and a vector

with nonnegative entries whose subtraction from  $\sum_{x \in A} c_x M(x, \cdot)$  gives  $\mathbf{t}$ . Reading this coordinate by coordinate gives the assertion.  $\square$

## 7 A set of vertices in place of one, and Conjecture 6

Conjecture 2.5 concerns a graph in which one pair  $e = \{v, w\}$  has been forced open, while the vertex  $a$  realising the minimum is prescribed in the original graph. For a finite set  $R$  of vertices write  $C_R = \bigcup_{s \in R} C_s$ , so that  $\{R \leftrightarrow B\}$  is the event  $\{C_R \cap B \neq \emptyset\}$  and  $\{R \nleftrightarrow B\}$  its complement. Nothing is assumed about the vertices of  $R$ , and the set  $C_R$  need not be connected. The following theorem is the form in which Conjecture 2.5 will be proved.

**Theorem 7.1.** *Let  $A$  be a nonempty finite set of vertices, let  $a \in A$ , let  $R$  be a nonempty finite set of vertices and let  $F$  be increasing with  $\mathbb{E}F(C_a) \leq \mathbb{E}F(C_x)$  for every  $x \in A$ . Then*

$$\mathbb{E}[F(C_R) - F(C_a); R \nleftrightarrow a, R \leftrightarrow A \setminus \{a\}] \geq 0. \quad (21)$$

Section 7.1 deduces Conjecture 2.5 from it, and the rest of the section proves it.

### 7.1 Forcing the pair open

**Lemma 7.2.** *Let  $v$  and  $w$  be vertices, put  $R = \{v, w\}$  and  $e = \{v, w\}$ , and let  $\omega$  be a configuration in which  $e$  is closed. Then  $C_R$  takes the same value in  $\omega$  and in the configuration obtained from  $\omega$  by opening  $e$ ; the same holds for the events  $\{R \nleftrightarrow B\}$  and  $\{R \leftrightarrow B\}$  for every set  $B$  of vertices, and, on  $\{R \nleftrightarrow a\}$ , for the cluster  $C_a$ .*

*Proof.* Opening  $e$  can only enlarge clusters, so  $C_R$  can only grow. Conversely, a path in the enlarged configuration from  $v$  or  $w$  to a vertex  $u$  that uses  $e$  splits at its uses of  $e$  into segments of  $\omega$ , each of which starts at  $v$  or at  $w$ ; the last segment exhibits  $u$  in  $C_v \cup C_w$  computed in  $\omega$ . Hence  $C_R$  is unchanged, and so are the events determined by  $C_R$ . If moreover  $R \nleftrightarrow a$  in  $\omega$  then no path of  $\omega$  from  $a$  meets  $\{v, w\}$ , so no path from  $a$  in the enlarged configuration can use  $e$ , and  $C_a$  is unchanged as well.  $\square$

**Lemma 7.3.** *Let  $\varphi$  be a real function of configurations that takes the same value on a configuration in which  $e$  is closed and on the configuration obtained from it by opening  $e$ . Then the mean of  $\varphi$  under  $\mathbb{P}^e$  equals its mean under  $\mathbb{P}$ .*

*Proof.* Condition on the states of all the pairs other than  $e$ . Under  $\mathbb{P}$  and under  $\mathbb{P}^e$  those states have the same joint law, since only the weight of  $e$  differs. Given them, the hypothesis says that  $\varphi$  takes one and the same value whether  $e$  is open or closed, so the two conditional means of  $\varphi$  agree, and averaging gives the assertion.  $\square$

**Theorem 7.4.** *Let  $A$  be a nonempty finite set of vertices, let  $b$  be a vertex, let  $a \in A$  minimise  $\mathbb{P}(x \leftrightarrow b)$  over  $x \in A$ , and let  $e = \{v, w\}$  be a pair. Then*

$$\mathbb{P}^e(a \leftrightarrow b, v \leftrightarrow A) \leq \mathbb{P}^e(v \leftrightarrow b, v \leftrightarrow A), \quad (22)$$

*and Conjecture 2.5 holds. Its hypothesis (2) is a consequence of Conjecture 2.1 and therefore carries no information.*

*Proof.* Apply Theorem 7.1 with  $R = \{v, w\}$  and  $F = \mathbf{1}\{b \in \cdot\}$ , for which  $\mathbb{E}F(C_x) = \mathbb{P}(x \leftrightarrow b)$ , so that the hypothesis on  $a$  is exactly the minimality assumed here. By Lemmas 7.2 and 7.3 the left side of (21) has the same value under  $\mathbb{P}$  and under  $\mathbb{P}^e$ , since the integrand and the two events defining the range of integration are unchanged by opening  $e$  on the range of integration. Hence

$$\mathbb{E}^e[F(C_R) - F(C_a); R \nleftrightarrow a, R \leftrightarrow A \setminus \{a\}] \geq 0, \quad R = \{v, w\}.$$

Under  $\mathbb{P}^e$  the pair  $e$  is open with probability one, so almost surely  $C_v = C_w = C_R$ , the event  $\{R \leftrightarrow a\}$  is  $\{v \leftrightarrow a\}$  and the event  $\{R \leftrightarrow A \setminus \{a\}\}$  is  $\{v \leftrightarrow A \setminus \{a\}\}$ . Splitting  $\{v \leftrightarrow A\}$  into  $\{v \leftrightarrow a\}$ , on which  $C_v = C_a$  and the integrand vanishes, and  $\{v \leftrightarrow a\} \cap \{v \leftrightarrow A \setminus \{a\}\}$ , the last display becomes

$$\mathbb{E}^e[\mathbf{1}\{b \in C_v\} - \mathbf{1}\{b \in C_a\}; v \leftrightarrow A] \geq 0,$$

which is (22). The events  $\{a \leftrightarrow b\}$  and  $\{v \leftrightarrow A\}$  are increasing, so the Harris inequality and (22) give

$$\mathbb{P}^e(v \leftrightarrow A)\mathbb{P}^e(a \leftrightarrow b) \leq \mathbb{P}^e(a \leftrightarrow b, v \leftrightarrow A) \leq \mathbb{P}^e(v \leftrightarrow b, v \leftrightarrow A) \leq \mathbb{P}^e(v \leftrightarrow b),$$

which is (3). Finally, Conjecture 2.1 applied with  $x$  in place of  $o$  gives  $\mathbb{P}(x \leftrightarrow b) \geq \mathbb{P}(x \leftrightarrow A) \min_{y \in A} \mathbb{P}(y \leftrightarrow b) = \mathbb{P}(x \leftrightarrow A)\mathbb{P}(a \leftrightarrow b)$ , which is (2).  $\square$

## 7.2 Events conditioned on a set missing another set

Throughout Sections 7.2 to 7.5 fix a vertex  $s$ , a set  $Y$  of vertices with  $s \notin Y$ , and an increasing function  $g$  of sets of pairs, and write  $D = \{s \leftrightarrow Y\}$ . For finite sets  $R, X$  and  $B$  of vertices put

$$H(R; X, B) = \{R \leftrightarrow B\} \cap \{R \leftrightarrow X\}, \quad (23)$$

the event that  $C_R$  misses  $B$  and meets  $X$ . The event  $\{R \leftrightarrow s\}$  does not imply  $\{R \leftrightarrow Y\}$  even when  $\{s \leftrightarrow Y\}$  holds, because a component of  $C_R$  other than the one containing  $s$  may meet  $Y$ , so the two conditions must be carried together; for a single vertex, however,  $H(\{u\}; \{s\}, Y) = D \cap \{s \leftrightarrow u\}$ , since  $u \leftrightarrow s$  makes the clusters of  $u$  and  $s$  equal. Set

$$\Gamma(R) = \mathbb{P}(D) \mathbb{E}[g(C_s); H(R; \{s\}, Y)] - \mathbb{E}[g(C_s); D] \mathbb{P}(H(R; \{s\}, Y)), \quad (24)$$

which at  $R = \{u\}$  is  $\mathbb{P}(D)^2$  times the conditioned covariance appearing in Theorem 3.2. The event  $H(R; \{s\}, Y)$  is contained in  $D$ : on it  $s$  lies in  $C_R$  and  $C_R$  misses  $Y$ , so  $C_s \subseteq C_R$  misses  $Y$ .

The functional (7) is now applied to functions of a set argument rather than of a vertex. Fix distinct vertices  $z_1, \dots, z_\ell$  and a further vertex  $v$ , all outside  $\{s\} \cup Y$  and distinct from each other, put  $B_1 = \{s\} \cup Y$  and  $B_{j+1} = B_j \cup \{z_j\}$ , and put

$$c_j(R) = \frac{\mathbb{P}(H(R; \{z_j\}, B_j))}{\mathbb{P}(z_j \leftrightarrow B_j)}, \quad \tau(R) = \frac{\mathbb{P}(H(R; \{v\}, B_{\ell+1}))}{\mathbb{P}(v \leftrightarrow B_{\ell+1})}. \quad (25)$$

For a real function  $q$  of finite vertex sets put  $q^0 = q$ , then  $q^j(R) = q^{j-1}(R) - c_j(R)q^{j-1}(\{z_j\})$  for  $1 \leq j \leq \ell$ , and finally, for a prescribed set  $R_0$ ,

$$M[q] = q^\ell(R_0) - \tau(R_0)q^\ell(\{v\}). \quad (26)$$

If  $R_0 = \{u\}$  for a vertex  $u$  outside  $B_{\ell+1}$  and distinct from  $v$ , then the numbers (25) at singletons are the constants (6) with  $o$  replaced by  $u$ , because  $\mathbb{P}(H(\{y\}; \{z\}, B)) = \mathbb{P}(z \leftrightarrow B, z \leftrightarrow y)$  for every  $y, z$  and  $B$ ; hence (26) is (7) and  $M[\Gamma] \geq 0$  is Theorem 3.2. The goal of Sections 7.2 to 7.5 is the same conclusion for a set  $R_0$  with at least two elements.

**Lemma 7.5.** *There are real numbers  $\lambda_u$  for  $u \in V$ , depending only on  $z_1, \dots, z_\ell, v$  and  $R_0$ , such that  $M[q] = q(R_0) + \sum_{u \in V} \lambda_u q(\{u\})$  for every real function  $q$  of finite vertex sets.*

*Proof.* Induct on  $\ell$ . For  $\ell = 0$  the assertion holds with  $\lambda_v = -\tau(R_0)$  and all other  $\lambda_u$  zero. For  $\ell \geq 1$  the functional (26) for the list  $z_1, \dots, z_\ell$  applied to  $q$  is the functional for  $z_2, \dots, z_\ell$  applied to  $R \mapsto q(R) - c_1(R)q(\{z_1\})$ . By the induction hypothesis the latter is that function at  $R_0$  plus a combination of its values at singletons; every term produced by the subtraction is a multiple of  $q(\{z_1\})$ , and the coefficient of  $q(R_0)$  stays one. Collecting equal vertices gives the assertion.  $\square$

Only the value at  $R_0$  enters with a coefficient not attached to a singleton, and that coefficient is one; this is what makes the one-sided estimates below usable.

### 7.3 Exploring the cluster of a set

Let  $z$  be a vertex, let  $B$  be a set of vertices containing  $\{s\} \cup Y$  but not  $z$ , and let  $\Theta$  and  $\Psi$  be as in (8) and the paragraph containing it, built from  $s$ ,  $Y$  and  $g$ . Abbreviate  $H_R = H(R; \{z\}, B)$  and  $E = \{z \leftrightarrow B\}$ , and assume  $\mathbb{P}(E) > 0$ .

**Lemma 7.6.** *For every finite set  $R$  of vertices,  $\mathbb{E}[\Theta; H_R] = -\mathbb{E}[\Psi(C_R); H_R]$ .*

*Proof.* Sum over the possible values of the complete set of open pairs met by paths from  $R$ . On  $H_R$  the vertex set  $C_R$  misses  $B$ , hence misses  $\{s\} \cup Y$ , and every pair with exactly one endpoint in  $C_R$  is closed. Given that data, the pairs with both endpoints outside  $C_R$  retain their original weights, and deleting every pair meeting  $C_R$  changes neither  $\mathcal{C}_s$  nor  $\mathcal{C}_Y$  nor the quantity  $\bar{g}$  of (8). The conditional mean of  $\Theta$  is therefore its mean in a configuration with every pair meeting  $C_R$  deleted, which is  $-\Psi(C_R)$ . The event  $H_R$  is determined by the same data, so summing over its values gives the assertion.  $\square$

**Lemma 7.7.** *Put  $\Delta(R) = \mathbb{E}[\Psi(C_R) - \Psi(C_z); H_R]$ . Then  $\Delta(R) \geq 0$  for every finite  $R$ , and  $\Delta(\{u\}) = 0$  for every vertex  $u$ .*

*Proof.* On  $H_R$  some vertex of  $R$  is joined to  $z$ , so  $C_z \subseteq C_R$  and  $\Psi(C_z) \leq \Psi(C_R)$  because  $\Psi$  is increasing. If  $R = \{u\}$  then on  $H_{\{u\}}$  the clusters of  $u$  and  $z$  coincide, so the integrand vanishes identically.  $\square$

The next inequality is the step that transfers the estimate from a single vertex to a set. Let  $\Gamma_{z,B}$  denote the expression (24) formed with the vertex  $z$ , the set  $B$  and the increasing function  $\mathcal{C} \mapsto \Psi(\{z\} \cup \text{span } \mathcal{C})$  in place of  $s$ ,  $Y$  and  $g$ .

**Lemma 7.8.** *For every finite set  $R$  of vertices,*

$$\mathbb{E}[\Theta; H_R] - c(R) \mathbb{E}[\Theta; E] = -\frac{\Gamma_{z,B}(R)}{\mathbb{P}(E)} - \Delta(R), \quad c(R) = \frac{\mathbb{P}(H_R)}{\mathbb{P}(E)}, \quad (27)$$

and consequently

$$\mathbb{E}[\Theta; H_R] - c(R) \mathbb{E}[\Theta; E] \leq -\frac{\Gamma_{z,B}(R)}{\mathbb{P}(E)}, \quad (28)$$

with equality in (28) whenever  $R$  is a single vertex.

*Proof.* Expanding (24) for the vertex  $z$ , the set  $B$  and the stated function, and using that  $\Psi(\{z\} \cup \text{span } \mathcal{C}_z) = \Psi(C_z)$ ,

$$\frac{\Gamma_{z,B}(R)}{\mathbb{P}(E)} = \mathbb{E}[\Psi(C_z); H_R] - c(R) \mathbb{E}[\Psi(C_z); E].$$

By Lemma 7.6 and the definition of  $\Delta$ ,  $\mathbb{E}[\Theta; H_R] = -\mathbb{E}[\Psi(C_z); H_R] - \Delta(R)$ , and by (9),  $\mathbb{E}[\Theta; E] = -\mathbb{E}[\Psi(C_z); E]$ . Substituting gives (27). The inequality (28) follows from  $\Delta \geq 0$ , with equality at a single vertex by Lemma 7.7.  $\square$

### 7.4 Transfer between two vertex sets in a restricted graph

Recall  $\mathbb{P}_U$ ,  $\mathbb{E}_U$  and  $\text{Cov}_U$  from Section 3. Every quantity below formed inside  $U$  is set to zero if a distinguished vertex appearing in it lies outside  $U$ , and every set appearing in it is intersected with  $U$ . Fix a set  $X$  of vertices containing  $s$ , a vertex  $v \notin X$ , and a finite set  $R$  of vertices disjoint from  $X \cup \{v\}$ . For a set  $K$  of vertices put

$$\alpha_{R,B}(K) = \mathbf{1}\{K \cap R \neq \emptyset\} \mathbb{P}_{U \setminus K}(R \setminus K \leftrightarrow B). \quad (29)$$

**Lemma 7.9.** *The function  $K \mapsto \alpha_{R,B}(K)$  is increasing, the function  $B \mapsto \alpha_{R,B}(K)$  is decreasing, and if  $\{v\} \cup R \cup B \subseteq U$  then*

$$\mathbb{E}_U[\alpha_{R,B}(C_v); v \leftrightarrow B] = \mathbb{P}_U(R \leftrightarrow B, R \leftrightarrow v). \quad (30)$$

*Proof.* Let  $K \subseteq K_1$ . The first factor can only turn from zero to one. For the second, couple the two measures so that the configuration inside  $U \setminus K_1$  is contained in the one inside  $U \setminus K$ ; since also  $R \setminus K_1 \subseteq R \setminus K$ , avoidance of  $B$  by  $R \setminus K$  inside  $U \setminus K$  forces avoidance of  $B$  by  $R \setminus K_1$  inside  $U \setminus K_1$ , so the second factor increases as well. Enlarging  $B$  makes avoidance harder and decreases the second factor. For (30), sum over the possible values of the complete set of open pairs met by paths from  $v$ . On  $\{v \leftrightarrow B\}$  the vertex set  $K = C_v$  misses  $B$ , the event  $\{R \leftrightarrow v\}$  is  $\{K \cap R \neq \emptyset\}$ , every pair with exactly one endpoint in  $K$  is closed, and the pairs with both endpoints outside  $K$  are fresh. On that data the event  $\{R \leftrightarrow B\}$  is the event that  $R \setminus K$  misses  $B$  in  $U \setminus K$ , since the vertices of  $R$  inside  $K$  have cluster  $C_v$ , which misses  $B$ .  $\square$

**Lemma 7.10.** *Let  $N$  and  $N_1$  be disjoint sets of vertices, and let all the sets and vertices named lie in  $U$ . Then*

$$\mathbb{P}_U(R \leftrightarrow X \cup N, R \leftrightarrow v) \mathbb{P}_U(v \leftrightarrow X \cup N_1) \leq \mathbb{P}_U(v \leftrightarrow X \cup N \cup N_1) \mathbb{P}_U(R \leftrightarrow X, R \leftrightarrow v). \quad (31)$$

*Proof.* By (30) the left side is  $\mathbb{E}_U[\alpha_{R, X \cup N}(C_v); v \leftrightarrow X \cup N] \mathbb{P}_U(v \leftrightarrow X \cup N_1)$ . Apply Theorem 3.1 inside  $U$  with the vertex  $v$ , the two sets  $X \cup N$  and  $X \cup N_1$ , whose intersection is  $X$  and whose union is  $X \cup N \cup N_1$  because  $N$  and  $N_1$  are disjoint, and the increasing functions  $\alpha_{R, X \cup N}$  and the constant one. The result is at most  $\mathbb{E}_U[\alpha_{R, X \cup N}(C_v); v \leftrightarrow X] \mathbb{P}_U(v \leftrightarrow X \cup N \cup N_1)$ , whose first factor is at most  $\mathbb{E}_U[\alpha_{R, X}(C_v); v \leftrightarrow X] = \mathbb{P}_U(R \leftrightarrow X, R \leftrightarrow v)$  by Lemma 7.9 and (30). If  $v$  lies in an avoided set or  $R$  meets one, the left side is zero.  $\square$

**Lemma 7.11.** *Let  $U \subseteq V$ , let  $h$  be a nonnegative real function of subsets of  $U$  that vanishes on sets not containing  $v$ , and for sets  $N$  and  $N_1$  of vertices contained in  $U$  put*

$$\phi(N) = \mathbb{E}_U[\mathbf{1}\{s \leftrightarrow N\} h(U \setminus C_N)], \quad \chi(N) = \mathbb{E}_U[\mathbf{1}\{s \leftrightarrow N\} \mathbf{1}\{R \leftrightarrow N\} \theta(U \setminus C_N) h(U \setminus C_N)],$$

where  $\theta(W) = \mathbb{P}_W(R \leftrightarrow X, R \leftrightarrow v) / \mathbb{P}_W(v \leftrightarrow X)$ . Assume  $s \in X$ ,  $v \notin X$ , that  $R$  is disjoint from  $X \cup \{v\}$ , that all the named sets and vertices lie in  $U$ , that in every  $W \subseteq U$  and for every  $Q \subseteq W$  one has  $\phi(Q) \mathbb{P}_W(v \leftrightarrow X) \leq \mathbb{P}_W(v \leftrightarrow X \cup Q) h(W)$  with  $\phi$  formed inside  $W$ , and that  $Q \mapsto \phi(Q)$  is decreasing in every such  $W$ . Then

$$\mathbb{P}_U(R \leftrightarrow X \cup N, R \leftrightarrow v) \phi(N_1) \leq \mathbb{P}_U(v \leftrightarrow X \cup N \cup N_1) \chi(N \cap N_1). \quad (32)$$

*Proof.* Induct on the number of elements of  $U$ . If  $N$  and  $N_1$  are disjoint, multiply the assumed inequality for  $N_1$  by  $\mathbb{P}_U(R \leftrightarrow X \cup N, R \leftrightarrow v)$  and apply Lemma 7.10, using  $\mathbb{P}_U(R \leftrightarrow X, R \leftrightarrow v) = \theta(U) \mathbb{P}_U(v \leftrightarrow X)$  and  $\chi(\emptyset) = \theta(U) h(U)$ . If  $\mathbb{P}_U(v \leftrightarrow X) = 0$  then  $\mathbb{P}_U(R \leftrightarrow X \cup N, R \leftrightarrow v) = 0$  by event inclusion and both sides vanish.

Otherwise put  $Z = N \cap N_1$ , which is nonempty. If  $v \in Z$ , or  $s \in Z$ , or  $R$  meets  $Z$ , the left side is zero. In the remaining case remove  $Z$  and expose the set  $\sigma \subseteq U \setminus Z$  of vertices joined to  $Z$  by an open pair. The indicators of the events  $\{u \in \sigma\}$  are determined by disjoint blocks of pairs, so  $\sigma$  has a product law  $\mathbb{Q}$  on subsets of  $U \setminus Z$ . Put  $N_2 = N \setminus Z$  and  $N_3 = N_1 \setminus Z$ , let  $\phi$  and  $\chi$  be formed inside  $U \setminus Z$ , and put

$$\kappa(\sigma) = \mathbb{P}_{U \setminus Z}(R \leftrightarrow X \cup N_2 \cup \sigma, R \leftrightarrow v), \quad \mu(\sigma) = \mathbb{P}_{U \setminus Z}(v \leftrightarrow X \cup N_2 \cup N_3 \cup \sigma).$$

The four quantities in (32) are the  $\mathbb{Q}$ -averages of  $\kappa(\sigma)$ , of  $\phi(N_3 \cup \sigma)$ , of  $\mu(\sigma)$  and of  $\chi(\sigma)$ : the event that  $R$  misses  $N$  in  $U$  is the event that  $R$  misses  $N_2 \cup \sigma$  in  $U \setminus Z$ , and likewise for the other three, so no factor is taken outside an expectation. For subsets  $\sigma_1$  and  $\sigma_2$  of  $U \setminus Z$  put  $N_4 = \sigma_1 \cup (N_2 \setminus \sigma_2)$  and  $N_5 = \sigma_2 \cup (N_3 \setminus \sigma_1)$ ; then  $N_4 \cup N_5 = N_2 \cup N_3 \cup \sigma_1 \cup \sigma_2$  and  $N_4 \cap N_5 = \sigma_1 \cap \sigma_2$ , so the induction hypothesis inside  $U \setminus Z$ , together with the assumed monotonicity of  $\phi$ , gives  $\kappa(\sigma_1) \phi(N_3 \cup \sigma_2) \leq \mu(\sigma_1 \cup \sigma_2) \chi(\sigma_1 \cap \sigma_2)$ . Multiplying by the product identity  $\mathbb{Q}(\sigma_1) \mathbb{Q}(\sigma_2) = \mathbb{Q}(\sigma_1 \cap \sigma_2) \mathbb{Q}(\sigma_1 \cup \sigma_2)$  and applying the Ahlswede and Daykin four functions theorem to the four functions  $\sigma \mapsto \mathbb{Q}(\sigma) \kappa(\sigma)$ ,  $\sigma \mapsto \mathbb{Q}(\sigma) \phi(N_3 \cup \sigma)$ ,  $\sigma \mapsto \mathbb{Q}(\sigma) \mu(\sigma)$  and  $\sigma \mapsto \mathbb{Q}(\sigma) \chi(\sigma)$  yields (32).  $\square$

**Lemma 7.12.** *Let  $s \in X$ , let  $v \notin X$ , let  $R$  be disjoint from  $X \cup \{v\}$  and let  $g$  be increasing. Put  $\eta_R(U) = \text{Cov}_U(g(\mathcal{C}_s), \mathbf{1}\{R \leftrightarrow X\})$ , and let  $\eta_v(U)$  be as in (10). Then  $\eta_v(U) \geq 0$  and*

$$\theta(U) \eta_v(U) \leq \eta_R(U). \quad (33)$$

*Proof.* All expectations are taken under  $\mathbb{P}_U$ , and the argument  $U$  is suppressed. If  $v \notin U$  then  $\eta_v = 0$  and the assertion is the Harris inequality. The events  $\{v \leftrightarrow X\}$  and  $\{g(\mathcal{C}_s) \geq t\}$  are increasing, so  $\eta_v \geq 0$  by the Harris inequality. Put  $J = \{R \leftrightarrow X\} \cap \{R \leftrightarrow v\}$  and  $D_1 = \{v \leftrightarrow X\}$ . If  $v \leftrightarrow X$  and  $R \leftrightarrow v$  then  $R \leftrightarrow X$ , so  $J \subseteq D_1$ , and  $\mathbf{1}\{R \leftrightarrow X\} + \mathbf{1}_J = \mathbf{1}\{R \leftrightarrow X \text{ or } R \leftrightarrow v\}$ , an increasing event. The Harris inequality applied to that event gives  $\eta_R + \text{Cov}_U(g(\mathcal{C}_s), \mathbf{1}_J) \geq 0$ , that is

$$\eta_R \geq \mathbb{E}_U[g(\mathcal{C}_s)] \mathbb{P}_U(J) - \mathbb{E}_U[g(\mathcal{C}_s); J].$$

Write  $\mathcal{C}_X$  and  $\mathcal{C}_v$  for the sets of open pairs lying on open paths from  $X$  and from  $v$ . The set  $\mathcal{C}_s$  is the connected component of  $\mathcal{C}_X$  containing  $s$ , so  $g(\mathcal{C}_s)$  is an increasing function of  $\mathcal{C}_X$  and does not depend on  $\mathcal{C}_v$ ; the event  $\{R \leftrightarrow X\}$  is  $\{R \cap (X \cup \text{span } \mathcal{C}_X) = \emptyset\}$  and the event  $\{R \leftrightarrow v\}$  is  $\{R \cap (\{v\} \cup \text{span } \mathcal{C}_v) \neq \emptyset\}$ , so  $\mathbf{1}_J$  is decreasing in  $\mathcal{C}_X$  and increasing in  $\mathcal{C}_v$ . The second part of Theorem 3.1, applied with  $S_1 = X$ ,  $S_2 = \{v\}$  and the two functions  $g(\mathcal{C}_s)$  and  $\mathbf{1}_J$ , gives  $\mathbb{P}_U(D_1) \mathbb{E}_U[g(\mathcal{C}_s); J] \leq \mathbb{P}_U(J) \mathbb{E}_U[g(\mathcal{C}_s); D_1]$ . Since  $\eta_v = \mathbb{E}_U[g(\mathcal{C}_s)] \mathbb{P}_U(D_1) - \mathbb{E}_U[g(\mathcal{C}_s); D_1]$  and  $\theta(U) = \mathbb{P}_U(J) / \mathbb{P}_U(D_1)$ ,

$$\theta(U) \eta_v = \mathbb{E}_U[g(\mathcal{C}_s)] \mathbb{P}_U(J) - \frac{\mathbb{P}_U(J) \mathbb{E}_U[g(\mathcal{C}_s); D_1]}{\mathbb{P}_U(D_1)} \leq \mathbb{E}_U[g(\mathcal{C}_s)] \mathbb{P}_U(J) - \mathbb{E}_U[g(\mathcal{C}_s); J] \leq \eta_R.$$

If  $\mathbb{P}_U(D_1) = 0$  then  $\eta_v = 0$  and the assertion is the Harris inequality.  $\square$

**Proposition 7.13.** *Let every pair weight lie strictly between zero and one, let  $X = \{s\} \cup \{z_1, \dots, z_\ell\}$ , let  $v \notin X \cup Y$ , let  $R_0$  be disjoint from  $X \cup Y \cup \{v\}$  and let  $g$  be increasing. Write  $U = V \setminus C_Y$  and let  $\eta_{R_0}$  and  $\eta_v$  be as in Lemma 7.12. Then*

$$\mathcal{H} := \mathbb{E}[\mathbf{1}_D \mathbf{1}\{R_0 \leftrightarrow Y\} \eta_{R_0}(U)] - \tau(R_0) \mathbb{E}[\mathbf{1}_D \eta_v(U)] \geq 0. \quad (34)$$

*Proof.* Subtracting from  $g$  its minimum over the finitely many values of  $\mathcal{C}_s$  changes neither  $\eta_{R_0}$  nor  $\eta_v$ , so assume  $g \geq 0$ . Take  $h = \eta_v$ , which is nonnegative by Lemma 7.12 and vanishes on sets not containing  $v$ ; the two hypotheses of Lemma 7.11 on  $h$  are (11) and the monotonicity recorded with it. Apply Lemma 7.11 with  $U = V$ , with  $R = R_0$  and with  $N = N_1 = Y$ , whose intersection and union are both  $Y$ . Its left side is  $\mathbb{P}(R_0 \leftrightarrow X \cup Y, R_0 \leftrightarrow v) \mathbb{E}[\mathbf{1}_D \eta_v(U)]$  and its right side is  $\mathbb{P}(v \leftrightarrow X \cup Y) \mathbb{E}[\mathbf{1}_D \mathbf{1}\{R_0 \leftrightarrow Y\} \theta(U) \eta_v(U)]$ . Dividing by  $\mathbb{P}(v \leftrightarrow X \cup Y)$ , which is positive because the configuration with no open pair has positive probability, and recognising  $\tau(R_0)$  as the resulting quotient by (25) with  $B_{\ell+1} = X \cup Y$ ,

$$\tau(R_0) \mathbb{E}[\mathbf{1}_D \eta_v(U)] \leq \mathbb{E}[\mathbf{1}_D \mathbf{1}\{R_0 \leftrightarrow Y\} \theta(U) \eta_v(U)].$$

By Lemma 7.12 applied inside  $U$ , the integrand on the right is at most  $\mathbf{1}_D \mathbf{1}\{R_0 \leftrightarrow Y\} \eta_{R_0}(U)$ .  $\square$

## 7.5 The inequality for a set of vertices

Recall  $\Theta$  from (8) and put

$$\mathcal{W} = \mathbb{E}[\Theta \cdot M[\beta]], \quad (35)$$

where  $\beta(R) = \mathbf{1}\{R \cap C_Y = \emptyset\} \mathbf{1}\{R \leftrightarrow s\}$  is the indicator of  $H(R; \{s\}, Y)$  and  $M$  acts on the argument  $R$  with the values of  $\mathcal{C}_s$  and  $\mathcal{C}_Y$  held fixed. Because  $\mathbb{E}[\Theta \mid \mathcal{C}_Y] = 0$ , the quantity  $\mathcal{W}$  is the average over  $\mathcal{C}_Y$  of a conditional covariance, and it is the part of  $M[\Gamma]$  produced after the cluster of  $Y$  has been exposed.

**Proposition 7.14.** *Let every pair weight lie strictly between zero and one, let  $z_1, \dots, z_\ell, v$  and  $R_0$  be as in Proposition 7.13, and let  $g$  be increasing. Then, with  $M_j$  the functional (26) built from  $z_{j+1}, \dots, z_\ell, v$  and  $R_0$  with base set  $B_{j+1}$ ,*

$$\mathcal{W} = \mathcal{H} + \sum_{j=1}^{\ell} \frac{M_j[\Gamma_{z_j, B_j}]}{\mathbb{P}(z_j \leftrightarrow B_j)} + \sum_{j=1}^{\ell} \Delta_j(R_0), \quad (36)$$

where  $\Gamma_{z_j, B_j}$  and  $\Delta_j$  are the quantities of Section 7.3 for the vertex  $z_j$  and the set  $B_j$ .

*Proof.* Write  $Z = C_Y$  and, for a set  $S$  of vertices and a vertex  $z$ ,

$$\beta_S(R) = \mathbf{1}\{R \cap Z = \emptyset, R \leftrightarrow S\}, \quad \delta_{S,z}(R) = \mathbf{1}\{R \cap Z = \emptyset, R \leftrightarrow S, R \leftrightarrow z\}.$$

Two pointwise identities hold:  $\beta_{S \cup \{z\}} = \beta_S + \delta_{S,z}$ , and  $\beta_S(\{z\}) = \mathbf{1}\{z \notin Z\} - \delta_{S,z}(\{z\})$ , the latter because connection is reflexive. Hence, for the operator  $q \mapsto q(\cdot) - c(\cdot)q(\{z\})$  with  $c$  as in (25),

$$\beta_S(R) - c(R)\beta_S(\{z\}) = \beta_{S \cup \{z\}}(R) - (\delta_{S,z}(R) - c(R)\delta_{S,z}(\{z\})) - c(R)\mathbf{1}\{z \notin Z\}.$$

Multiply by  $\Theta$  and take expectations. Every later operator and the final subtraction have constant coefficients, so the last term contributes a constant multiple of  $\mathbb{E}[\Theta \mathbf{1}\{z \notin Z\}]$ , which vanishes because  $\mathbf{1}\{z \notin Z\}$  is a function of  $\mathcal{C}_Y$  and  $\mathbb{E}[\Theta \mid \mathcal{C}_Y] = 0$ . With  $S = \{s\} \cup \{z_1, \dots, z_{j-1}\}$  and  $z = z_j$ , the event  $\{R \cap Z = \emptyset, R \leftrightarrow S\}$  is  $\{R \leftrightarrow B_j\}$ , and  $\delta_{S,z_j}(\{z_j\})$  is the indicator of  $\{z_j \leftrightarrow B_j\}$ ; the middle bracket is therefore the left side of (27) for the vertex  $z_j$  and the set  $B_j$ . Applying (27),

$$\mathbb{E}[\Theta(\beta_S(R) - c(R)\beta_S(\{z_j\}))] = \mathbb{E}[\Theta \beta_{S \cup \{z_j\}}(R)] + \frac{\Gamma_{z_j, B_j}(R)}{\mathbb{P}(z_j \leftrightarrow B_j)} + \Delta_j(R).$$

Start from  $\beta_{\{s\}}$ , which is the  $\beta$  of (35), iterate from  $j = 1$  to  $j = \ell$  and then subtract  $\tau(R_0)$  times the value at  $\{v\}$ . The terms produced at step  $j$  are acted on only by the operators of the vertices after  $z_j$  and by the final subtraction, which together form  $M_j$ ; by Lemma 7.5 applied to  $M_j$  they are evaluated at  $R_0$  with coefficient one and at singletons otherwise, and  $\Delta_j$  vanishes at every singleton by Lemma 7.7, so only  $\Delta_j(R_0)$  survives among those terms. After all  $\ell$  steps the remaining test function is  $\beta_X$  with  $X = \{s\} \cup \{z_1, \dots, z_\ell\}$ , and

$$\mathbb{E}[\Theta(\beta_X(R_0) - \tau(R_0)\beta_X(\{v\}))] = \mathcal{H},$$

because conditioning on  $\mathcal{C}_Y$  and using  $\mathbb{E}[\Theta \mid \mathcal{C}_Y] = 0$  turns each of the two terms into the conditional covariance of  $g(\mathcal{C}_s)$  with the corresponding connection indicator inside  $U = V \setminus C_Y$ , which are  $\eta_{R_0}(U)$  and  $\eta_v(U)$  of Lemma 7.12, carrying the factors  $\mathbf{1}_D \mathbf{1}\{R_0 \leftrightarrow Y\}$  and  $\mathbf{1}_D$ , the latter because  $\eta_v(U)$  vanishes when  $v \notin U$ . This is (36).  $\square$

**Proposition 7.15.** *Let every pair weight lie strictly between zero and one, let  $R_0$  be a finite set of vertices with at least two elements, and let  $z_1, \dots, z_\ell, v, s$  and  $Y$  satisfy the disjointness conditions of Section 7.2 with  $R_0$  disjoint from  $\{s, v\} \cup Y \cup \{z_1, \dots, z_\ell\}$ . Then  $M[\Gamma] \geq 0$  for every increasing  $g$ .*

*Proof.* Induct on  $\ell$ , the statement being asserted at each  $\ell$  for all admissible data simultaneously, and prove at each step both that  $\mathcal{W} \geq 0$  for every increasing  $g$  and that  $M[\Gamma] \geq 0$  for every increasing  $g$ .

For the first assertion, replace  $g$  by  $g$  minus its minimum, apply Proposition 7.14 and estimate the three groups of terms in (36). The first is nonnegative by Proposition 7.13. Each denominator  $\mathbb{P}(z_j \leftrightarrow B_j)$  is positive because  $z_j \notin B_j$  and the configuration with no open pair has positive probability, each  $M_j[\Gamma_{z_j, B_j}]$  is a quantity of the same shape with a strictly shorter list and with the nonnegative increasing  $\Psi$  in place of  $g$ , hence nonnegative by the induction hypothesis, and each  $\Delta_j(R_0)$  is nonnegative by Lemma 7.7. Hence  $\mathcal{W} \geq 0$ .

For the second assertion, expand  $\Gamma$ . By (24) and the containment  $H(R; \{s\}, Y) \subseteq D$ , and writing the two indicators of  $H(R; \{s\}, Y)$  in terms of the vertex carriers  $\{s\} \cup \text{span } \mathcal{C}_s$  and  $Y \cup \text{span } \mathcal{C}_Y$ ,

$$\Gamma(R) = \mathbb{P}(D) \mathbb{E}[g(\mathcal{C}_s) \beta_*(R); D] - \mathbb{E}[g(\mathcal{C}_s); D] \mathbb{E}[\beta_*(R); D],$$

in which  $\beta_*(R)$  is the indicator that  $R$  meets  $\{s\} \cup \text{span } \mathcal{C}_s$  and misses  $Y \cup \text{span } \mathcal{C}_Y$ . Since  $M$  is linear in its argument and its coefficients are constants,  $M[\Gamma]$  is the same expression with  $\beta_*(R)$  replaced by  $M[\beta_*]$ , a function of the two sets  $\mathcal{C}_s$  and  $\mathcal{C}_Y$ . Modify  $M[\beta_*]$  off the event  $D$  by dropping the second indicator at every singleton evaluation; on  $D$  the two clusters are disjoint and nothing changes, while the modified function is decreasing in  $\mathcal{C}_Y$  because by Lemma 7.5 only the evaluation at  $R_0$  depends on  $\mathcal{C}_Y$  and it enters with coefficient one. Here the hypothesis that  $R_0$  has at least two elements is used, since  $R_0$  must not itself be a singleton.

For real functions  $\varphi$  and  $\psi$  of configurations write  $\text{Cov}_D(\varphi, \psi) = \mathbb{E}[\varphi\psi; D] - \mathbb{E}[\varphi; D]\mathbb{E}[\psi; D]/\mathbb{P}(D)$ , so that the last paragraph reads  $M[\Gamma] = \mathbb{P}(D) \text{Cov}_D(g(\mathcal{C}_s), \psi)$  with  $\psi$  the modified function of  $(\mathcal{C}_s, \mathcal{C}_Y)$ . Let  $\varphi$  be a function of  $\mathcal{C}_s$  and put  $\bar{\varphi} = \mathbb{E}[\varphi \mid \mathcal{C}_Y]$ . Since  $D$  is determined by  $\mathcal{C}_Y$  and  $\mathbb{E}[\bar{\varphi}; D] = \mathbb{E}[\varphi; D]$ ,

$$\text{Cov}_D(\varphi, \psi) = \mathbb{E}[\mathbf{1}_D(\varphi - \bar{\varphi})\psi] + \mathbb{E}[\mathbf{1}_D(\bar{\varphi} - \mathbb{E}[\bar{\varphi} \mid \mathcal{C}_s])\psi] + \text{Cov}_D(\mathbb{E}[\bar{\varphi} \mid \mathcal{C}_s], \psi).$$

For  $\varphi = g(\mathcal{C}_s)$  the first term is  $\mathcal{W}$ , nonnegative for every increasing  $g$  by the first half of the proof. The second term is  $\mathbb{E}[\mathbf{1}_D \text{Cov}(\bar{\varphi}, \psi \mid \mathcal{C}_s)]$ ; given  $\mathcal{C}_s$  and on  $D$  the configuration outside  $\mathcal{C}_s$  is fresh, and both  $\bar{\varphi}$  and  $\psi$  are decreasing in  $\mathcal{C}_Y$ , hence in that configuration, so the conditional covariance is nonnegative by the Harris inequality. The third term has the same shape with  $\varphi$  replaced by  $\mathbb{E}[\bar{\varphi} \mid \mathcal{C}_s]$ , again an increasing function of  $\mathcal{C}_s$  because  $\bar{\varphi}$  is a decreasing function of  $\mathcal{C}_Y$ . Iterating, the third term becomes  $\text{Cov}_D(\varphi_n, \psi)$  where  $\varphi_n$  is the  $n$ th iterate of the averaging operator  $\varphi \mapsto \mathbb{E}[\mathbb{E}[\varphi \mid \mathcal{C}_Y] \mid \mathcal{C}_s]$ . Every configuration has positive probability, so that operator is given by a stochastic matrix with all entries positive on the finitely many values of  $\mathcal{C}_s$ ; the oscillation of  $\varphi_n$  therefore tends to zero, and  $\text{Cov}_D(\varphi_n, \psi) \rightarrow 0$  because  $\text{Cov}_D$  vanishes on constants. Hence  $M[\Gamma] \geq 0$ .  $\square$

## 7.6 Removing the vertices of the set one at a time

Fix a set  $Y$ , an increasing function  $F$  of vertex sets, a finite  $T \subseteq V \setminus Y$  with  $\mathbb{P}(x \leftrightarrow Y) > 0$  for  $x \in T$ , and an injective  $r$  on  $T$  with  $m_x \leq m_y$  whenever  $r(x) < r(y)$ , the numbers  $m_x$  being those of (12). For a finite set  $R$  of vertices define, in analogy with (13) and (14),

$$J_x(R) = \{R \leftrightarrow Y\} \cap \{R \leftrightarrow x\} \cap \bigcap_{y \in T, r(y) < r(x)} \{R \leftrightarrow y\},$$

$$\Lambda_R(T) = \mathbb{E}[F(C_R); R \leftrightarrow Y, R \leftrightarrow T] - \sum_{x \in T} \mathbb{P}(J_x(R)) m_x.$$

For  $R = \{u\}$  these are (13) and (14).

**Theorem 7.16.** *Let  $V$  be a finite vertex set with pair weights in  $[0, 1]$ , let  $Y$ ,  $F$ ,  $T$  and  $r$  be as above, and let  $R$  be a nonempty finite set of vertices disjoint from  $Y \cup T$ . Then  $\Lambda_R(T) \geq 0$ .*

*Proof.* Assume first that every pair weight lies strictly between zero and one. If  $R$  is a single vertex the assertion is Theorem 4.1, so let  $R$  have at least two elements.

Three preliminary facts are needed. First, let  $k \notin T$  satisfy  $r(y) < r(k)$  for every  $y \in T$ , and put  $E = \{k \leftrightarrow Y \cup T\}$  and  $H_R = H(R; \{k\}, Y \cup T)$ . Adding  $k$  leaves the events  $J_x(R)$  for  $x \in T$  unchanged and creates the single new event  $H_R$ , so

$$\Lambda_R(T \cup \{k\}) - \Lambda_R(T) = \mathbb{E}[F(C_R) - m_k; H_R], \quad (37)$$

and on  $H_R$  some vertex of  $R$  is joined to  $k$ , so  $C_k \subseteq C_R$  and the right side is at least  $\mathbb{E}[F(C_k) - m_k; H_R]$ , with equality when  $R$  is a single vertex. Second, with  $\gamma_k = m_k \mathbb{P}(E) - \mathbb{E}[F(C_k); E]$  and with  $\Gamma_{k, Y \cup T}$  the quantity (24) formed with the vertex  $k$ , the set  $Y \cup T$  and the increasing function  $\mathcal{C} \mapsto F(\{k\} \cup \text{span } \mathcal{C})$  in the roles of  $s$ ,  $Y$  and  $g$ ,

$$\mathbb{P}(E) \mathbb{E}[F(C_k) - m_k; H_R] = \Gamma_{k, Y \cup T}(R) - \gamma_k \mathbb{P}(H_R), \quad (38)$$



as one sees by expanding both sides. Third, exactly as in the proof of Proposition 4.2,

$$\gamma_k \leq \Lambda_{\{k\}}(T). \quad (39)$$

Let now  $M$  be the functional (26) with  $R_0 = R$ , with the vertices  $z_1, \dots, z_\ell$  and  $v$  outside  $Y \cup T \cup R$ , and with the sets (25) based at  $Y \cup T$ . We claim that

$$M[S \mapsto \Lambda_S(T)] \geq 0 \quad (40)$$

for every such choice, by induction on the cardinality of  $T$ . For  $T = \emptyset$  every value vanishes. Otherwise let  $k \in T$  have the largest value of  $r$ , put  $T_0 = T \setminus \{k\}$ , and read (37) for  $T_0$  and  $k$ . By Lemma 7.5 the value at  $R$  enters  $M$  with coefficient one and every other evaluation is at a singleton, where (37) is an equality; hence (37) may be inserted inside  $M$  with the correct sign. Multiplying by  $\mathbb{P}(E) > 0$  and using (38),

$$\mathbb{P}(E) M[\Lambda.(T)] \geq \mathbb{P}(E) M[\Lambda.(T_0)] + M[\Gamma_{k,Y \cup T_0}] - \gamma_k \mathbb{P}(E) M[c],$$

where  $c(S) = \mathbb{P}(H(S; \{k\}, Y \cup T_0)) / \mathbb{P}(E)$ . The middle term is nonnegative by Proposition 7.15 applied to the vertex  $k$ , the set  $Y \cup T_0$  and the increasing function  $\mathcal{C} \mapsto F(\{k\} \cup \text{span } \mathcal{C})$ . The last margin is nonnegative: Proposition 7.15 applied to the same vertex and set and to the increasing function  $\mathcal{C} \mapsto \mathbf{1}\{\mathcal{C} \neq \emptyset\}$  gives  $M[\Gamma_{k,Y \cup T_0}^*] \geq 0$  for the corresponding quantity  $\Gamma^*$ , and a direct expansion gives  $\Gamma^*(S) = \rho \mathbb{P}(H(S; \{k\}, Y \cup T_0))$  with  $\rho$  the probability that  $k$  misses  $Y \cup T_0$  and lies in no open pair. Indeed  $M$  evaluates its argument at  $R$  and at the singletons  $\{z_1\}, \dots, \{z_\ell\}, \{v\}$ , none of which contains  $k$ , so on  $H(S; \{k\}, Y \cup T_0)$  the vertex  $k$  is joined to a vertex other than itself and therefore lies in an open pair. The number  $\rho$  is positive because the configuration with no open pair has positive probability, so dividing by  $\rho \mathbb{P}(E)$  gives  $M[c] \geq 0$ . By (39) and  $M[c] \geq 0$ ,

$$M[\Lambda.(T_0)] - \gamma_k M[c] \geq M[\Lambda.(T_0)] - \Lambda_{\{k\}}(T_0) M[c],$$

and by the recursion defining (26) the right side is the corresponding quantity for the set  $T_0$  and the list obtained by prepending  $k$  to  $z_1, \dots, z_\ell$ , which is nonnegative by the induction hypothesis. Dividing by  $\mathbb{P}(E)$  gives (40).

Taking the set  $T_0$  in place of  $T$ ,  $\ell = 0$  and  $v = k$  in (40) gives

$$\mathbb{P}(H(R; \{k\}, Y \cup T_0)) \Lambda_{\{k\}}(T_0) \leq \mathbb{P}(k \leftrightarrow Y \cup T_0) \Lambda_R(T_0) \quad (41)$$

for every  $k$  outside  $Y \cup T_0 \cup R$ . We now prove  $\Lambda_R(T) \geq 0$  by induction on the cardinality of  $T$ . For  $T = \emptyset$  the quantity vanishes. Otherwise let  $k \in T$  have the largest value of  $r$ , put  $T_0 = T \setminus \{k\}$ ,  $E = \{k \leftrightarrow Y \cup T_0\}$  and  $H_R = H(R; \{k\}, Y \cup T_0)$ , so that (37) reads  $\Lambda_R(T) = \Lambda_R(T_0) + \mathbb{E}[F(C_R) - m_k; H_R]$ . On  $H_R$  we have  $C_k \subseteq C_R$ , so

$$\mathbb{E}[F(C_R) - m_k; H_R] \geq \mathbb{E}[F(C_k) - m_k; H_R].$$

Conditionally on the complete set of open pairs met by paths from  $k$ , and on  $E$ , the conditional probability of  $H_R$  is  $\alpha_{R,Y \cup T_0}(C_k)$  with  $\alpha$  as in (29) and  $U = V$ , an increasing function of that data by Lemma 7.9, and  $\mathbb{E}[\alpha_{R,Y \cup T_0}(C_k); E] = \mathbb{P}(H_R)$  by (30). Hence (5) applied with  $s = k$ , with the set  $Y \cup T_0$  and with the two increasing functions  $\mathcal{C} \mapsto F(\{k\} \cup \text{span } \mathcal{C})$  and  $\mathcal{C} \mapsto \alpha_{R,Y \cup T_0}(\{k\} \cup \text{span } \mathcal{C})$  gives  $\mathbb{P}(E) \mathbb{E}[F(C_k); H_R] \geq \mathbb{P}(H_R) \mathbb{E}[F(C_k); E]$ , whence

$$\mathbb{P}(E) \mathbb{E}[F(C_k) - m_k; H_R] \geq -\mathbb{P}(H_R) \gamma_k.$$

Multiplying the peeling identity by  $\mathbb{P}(E)$  and using (41), (39) and the last display,

$$\mathbb{P}(E) \Lambda_R(T) \geq \mathbb{P}(H_R) \Lambda_{\{k\}}(T_0) - \mathbb{P}(H_R) \gamma_k \geq 0,$$

and dividing by  $\mathbb{P}(E) > 0$  finishes the induction.

It remains to pass to weights in  $[0, 1]$ . As in (15), the assumption on  $r$  gives

$$\Lambda_R(T) = \mathbb{E}\left[\mathbf{1}\{R \leftrightarrow Y, R \leftrightarrow T\} \left(F(C_R) - \min_{x \in T \cap C_R} m_x\right)\right],$$

an expression not referring to  $r$  and, by the argument given after (15), continuous at every weight vector at which  $\mathbb{P}(x \leftrightarrow Y) > 0$  for each  $x \in T$ . Approximate such a vector by vectors with all coordinates strictly inside the unit interval and choose for each an injective  $r$  ordered by its own numbers  $m_x$ ; the case already proved applies to each of them, and continuity gives the assertion.  $\square$

## 7.7 Proof of Theorem 7.1

Two identities relate  $F$  to the function  $\Phi_a$  of Section 5 when a set of vertices replaces a single one.

**Lemma 7.17.** *Let  $R$  be a nonempty finite set of vertices, let  $a$  be a vertex, let  $F$  be increasing and let  $\mathcal{K}$  denote the complete set of open pairs met by paths from  $R$ . Then for every family  $\mathcal{S}$  of sets of pairs,*

$$\mathbb{E}[F(C_R) - F(C_a); R \leftrightarrow a, \mathcal{K} \in \mathcal{S}] = \mathbb{E}[\Phi_a(C_R); R \leftrightarrow a, \mathcal{K} \in \mathcal{S}].$$

*If moreover some  $s \in R$  satisfies  $\mathbb{E}F(C_a) \leq \mathbb{E}F(C_s)$ , then  $\mathbb{E}[F(C_R) - F(C_a); R \leftrightarrow a] \geq 0$ .*

*Proof.* For the identity, sum over the values of  $\mathcal{K}$ . On  $\{R \leftrightarrow a\}$  the vertex  $a$  lies outside  $C_R$ , every pair with exactly one endpoint in  $C_R$  is closed and the pairs with both endpoints outside  $C_R$  are fresh, so the conditional mean of  $F(C_a)$  is  $F(C_R) - \Phi_a(C_R)$ .

For the inequality, assume  $a \neq s$ , since otherwise  $\{R \leftrightarrow a\}$  is empty by reflexivity. Let  $\pi(K)$  be the probability that no vertex of  $R \setminus \{s\}$  reaches  $a$  in a fresh configuration with every pair meeting  $\{s\} \cup \text{span } K$  deleted. Enlarging  $K$  deletes more pairs, so  $\pi$  is nonnegative and increasing. Summing over the values of  $\mathcal{C}_s$  gives, for every real function  $p$  of sets of pairs,

$$\mathbb{E}[p(\mathcal{C}_s); R \leftrightarrow a] = \mathbb{E}[p(\mathcal{C}_s)\pi(\mathcal{C}_s); s \leftrightarrow a],$$

because on  $\{s \leftrightarrow a\}$  a vertex of  $R$  inside  $C_s$  misses  $a$  automatically, while for a vertex outside  $C_s$  both its cluster and  $C_a$  are computed in the fresh configuration. Apply (5) with the vertex  $s$ , with  $X = \{a\}$  and with the increasing functions  $p(\mathcal{C}) = \Phi_a(\{s\} \cup \text{span } \mathcal{C})$  and  $\pi$ :

$$\mathbb{E}[\Phi_a(C_s); s \leftrightarrow a] \mathbb{E}[\pi(C_s); s \leftrightarrow a] \leq \mathbb{P}(s \leftrightarrow a) \mathbb{E}[\Phi_a(C_s); R \leftrightarrow a].$$

The first factor on the left is  $\mathbb{E}F(C_s) - \mathbb{E}F(C_a) \geq 0$  by Lemma 5.1(ii) and the second is nonnegative, so if  $\mathbb{P}(s \leftrightarrow a) > 0$  then  $\mathbb{E}[\Phi_a(C_s); R \leftrightarrow a] \geq 0$ , while if  $\mathbb{P}(s \leftrightarrow a) = 0$  the event  $\{R \leftrightarrow a\}$  is null. Since  $\Phi_a$  is increasing and  $C_s \subseteq C_R$ , the identity of the first part with  $\mathcal{S}$  containing every set of pairs gives  $\mathbb{E}[F(C_R) - F(C_a); R \leftrightarrow a] = \mathbb{E}[\Phi_a(C_R); R \leftrightarrow a] \geq \mathbb{E}[\Phi_a(C_s); R \leftrightarrow a] \geq 0$ .  $\square$

*Proof of Theorem 7.1.* Put  $T = A \setminus \{a\}$ . If  $a \in R$  then  $\{R \leftrightarrow a\}$  is empty by reflexivity and both sides of (21) vanish. If  $R$  meets  $T$ , choose  $s \in R \cap T$ ; then  $\{R \leftrightarrow T\}$  is the sure event, so the left side of (21) is  $\mathbb{E}[F(C_R) - F(C_a); R \leftrightarrow a]$ , and since  $s \in A$  the hypothesis gives  $\mathbb{E}F(C_a) \leq \mathbb{E}F(C_s)$ , so Lemma 7.17 applies. In the remaining case  $R$  is disjoint from  $A$ .

Let  $T_1$  be the set of  $x \in T$  with  $\mathbb{P}(x \leftrightarrow a) > 0$ . Apply Theorem 7.16 with  $Y = \{a\}$ , with the set  $T_1$ , with the increasing function  $\Phi_a$  and with the set  $R$ , which is disjoint from  $\{a\} \cup T_1$ . By Lemma 5.1(ii) the number (12) attached to  $x \in T_1$  is  $(\mathbb{E}F(C_x) - \mathbb{E}F(C_a))/\mathbb{P}(x \leftrightarrow a) \geq 0$ , so the subtracted sum is nonnegative and

$$\mathbb{E}[\Phi_a(C_R); R \leftrightarrow a, R \leftrightarrow T_1] \geq 0.$$

The event  $\{R \leftrightarrow T_1\}$  is determined by the complete set of open pairs met by paths from  $R$ , so the first part of Lemma 7.17 converts this into  $\mathbb{E}[F(C_R) - F(C_a); R \leftrightarrow a, R \leftrightarrow T_1] \geq 0$ . Finally, a configuration in  $\{R \leftrightarrow a\} \cap \{R \leftrightarrow T\}$  but not in  $\{R \leftrightarrow T_1\}$  has  $R$  joined to some  $x \in T \setminus T_1$  while missing  $a$ , hence lies in the null event  $\{x \leftrightarrow a\}$ ; a finite union of null events is null, so the two set integrals agree and (21) follows.  $\square$

## 8 Deleting the edges at the starting vertex

Question 2.9 selects the vertex  $a$  by a probability computed with every pair incident to  $o$  closed, and then asks for (4) in the original measure. Exposing the pairs at  $o$  turns the cluster of  $o$  into the union of the clusters of a random set of vertices, and Theorem 7.1 applies to each value of that set.

**Theorem 8.1.** *Let  $A$  be a nonempty finite set of vertices, let  $o$  and  $b$  be vertices, and let  $a \in A$  minimise  $\mathbb{P}_\star(x \leftrightarrow b)$  over  $x \in A$ , where  $\mathbb{P}_\star$  is the percolation measure in which every pair incident to  $o$  is closed and all other weights are unchanged. Then (4) holds. In particular Question 2.9 has an affirmative answer.*

*Proof.* Condition on the states of the pairs incident to  $o$ . For such a state  $\zeta$  let  $R_\zeta$  be the union of  $\{o\}$  and the set of vertices  $u \neq o$  with  $\{o, u\}$  open in  $\zeta$ . The pairs not incident to  $o$  have exactly their law under  $\mathbb{P}_\star$ , and they are independent of  $\zeta$ . Under  $\mathbb{P}$  a path leaving  $o$  starts with a pair of  $\zeta$  and continues among the pairs not incident to  $o$ , possibly returning to  $o$  and leaving again, so, pointwise in the remaining configuration,

$$C_o = C_{R_\zeta}, \quad \{o \leftrightarrow A\} = \{R_\zeta \leftrightarrow A\}, \quad (42)$$

the right sides being computed under  $\mathbb{P}_\star$ , for which  $o$  is isolated and therefore contributes the single vertex  $o$  to the union. Moreover, on  $\{o \leftrightarrow a\}$  no path from  $a$  can reach  $o$ , so no such path uses a pair of  $\zeta$ , and  $C_a$  is the same under both measures.

Put  $F = \mathbf{1}\{b \in \cdot\}$ . On  $\{o \leftrightarrow a\}$  the clusters of  $o$  and  $a$  coincide, so

$$\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) - \mathbb{P}(a \leftrightarrow b, o \leftrightarrow A) = \mathbb{E}[F(C_o) - F(C_a); o \leftrightarrow a, o \leftrightarrow A \setminus \{a\}],$$

and by (42) the right side equals

$$\sum_{\zeta} \mathbb{P}(\zeta) \mathbb{E}_\star[F(C_{R_\zeta}) - F(C_a); R_\zeta \leftrightarrow a, R_\zeta \leftrightarrow A \setminus \{a\}],$$

the events and clusters inside the expectation being computed under  $\mathbb{P}_\star$ . The hypothesis on  $a$  says  $\mathbb{E}_\star F(C_a) \leq \mathbb{E}_\star F(C_x)$  for every  $x \in A$ , so each summand is nonnegative by Theorem 7.1 applied under  $\mathbb{P}_\star$  with the set  $R_\zeta$ . Averaging gives (4). No division is used, so states  $\zeta$  of probability zero and coincidences among  $o$ ,  $a$ ,  $b$  and the vertices of  $A$  are covered.  $\square$

## 9 A four-vertex example

Question 2.8 selects the vertex  $a$  by the probability of the joint event that  $a$  reaches  $b$  and that  $o$  does not reach  $A$ . When several vertices of  $A$  realise the minimum, the reading in which every one of them satisfies (4) is false.

**Proposition 9.1.** *There are a graph with four vertices, weights in  $[0, 1]$ , a set  $A$  with two elements and vertices  $o$  and  $b$  outside  $A$  such that both elements of  $A$  minimise  $\mathbb{P}(x \leftrightarrow b, o \leftrightarrow A)$  over  $x \in A$  while (4) fails for one of them.*

*Proof.* Let  $V = \{o, a_1, a_2, b\}$  and  $A = \{a_1, a_2\}$ , give the pairs  $\{o, a_1\}$  and  $\{a_2, b\}$  weight one and the pair  $\{a_1, a_2\}$  weight one half, and give every other pair weight zero. Since  $\{o, a_1\}$  is open with probability one and  $a_1 \in A$ , the event  $\{o \leftrightarrow A\}$  has probability one, so  $\mathbb{P}(x \leftrightarrow b, o \leftrightarrow A) = 0$  for both  $x \in A$  and both vertices of  $A$  minimise. The event  $\{a_2 \leftrightarrow b\}$  has probability one, so

$$\mathbb{P}(o \leftrightarrow A, a_2 \leftrightarrow b) = 1.$$

The event  $\{o \leftrightarrow b\}$  requires the pair  $\{a_1, a_2\}$  to be open, so  $\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) = \frac{1}{2}$ . Thus (4) fails for  $a = a_2$ .  $\square$

The minimised quantity vanishes at both vertices of  $A$  in this example. If  $a$  is required to be the unique minimiser, the question is open. If  $a$  is chosen to minimise  $\mathbb{P}(o \leftrightarrow A, x \leftrightarrow b)$  rather than  $\mathbb{P}(x \leftrightarrow b, o \leftrightarrow A)$ , then (4) holds by (1).

## 10 The Lean formalization

The results above are formalized in Lean 4 with Mathlib, on top of the formalization of Theorem 3.2, under Lean 4.32.0 with Mathlib commit 81a5d257c8e410db227a6665ed08f64fea08e997; every theorem named here compiles with no `sorry` and depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. Theorems 4.1 and 5.2 are `KNA11.genY_all` and `KNA11.conjecture4Fixed_holds`; Conjectures 2.4, 2.2, 2.1 and 2.3 and Question 2.7 are `KNA11.conjecture4_holds`, `KNA11.conjecture2Strong_holds`, `KNA11.conjecture2_holds`, `KNA11.conjecture1_holds`, `KNA11.conjecture3_holds` and `KNA11.question7_holds`, in `KN/Conjectures.lean` and `KN/ClusterProperty.lean`; Theorem 6.2 is `KNA11.question5_holds`; Theorem 7.1 for a set of two vertices and Theorem 7.4 are `KNA11.Guarded.pair_fixedMin`, `KNA11.Guarded.conjecture6Strong_holds` and `KNA11.Guarded.conjecture6_holds`, in the modules `GuardedDefs`, `GuardedBasic`, `GuardedKernel`, `GuardedDecoy`, `GuardedTwoCluster`, `PairGuardedCSH`, `PairSurplus`, `PairSurplusClosure`, `PairFixedMin` and `Conjecture6Proof`; the reduction of Theorem 8.1 is `KNA11.question9_of_setSourceFixedMin`; and Proposition 9.1 is `KNA11.not.question8EveryMin`. The modules for Theorem 7.1 with an arbitrary finite set, which Theorem 8.1 uses, follow the proofs given above with a set in place of a pair.

All of this was done autonomously by ChatGPT 5.6 Sol and Claude Fable 5.1, with minimal human intervention.

## References

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- [5] G. Kozma and S. Nitzan, *A reduction of the percolation at criticality problem to a conjectured inequality*, arXiv:2401.12397, 2024. Conjectures 1 to 3 with display (3) are on pp. 3 and 15, Conjecture 4 and Question 5 on pp. 31–32, Conjecture 6 on p. 34, and Questions 7 to 9 with display (41) on p. 36.